

# 3D Gaussian Splatting: A Comprehensive Survey of Representations, Methodologies, and Emerging Frontiers

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# 1 Introduction

The pursuit of photorealistic, interactive three-dimensional (3D) reconstruction and novel view synthesis has been a central ambition in computer graphics and vision, marked by a continuous oscillation between representational fidelity and computational tractability. The field’s trajectory reveals a progression from explicit, discrete geometric primitives—meshes and point clouds—towards implicit, coordinate-based neural networks that achieve unprecedented visual quality but at a prohibitive computational cost. The introduction of Neural Radiance Fields (NeRFs) marked a watershed moment, offering a remarkable leap in photorealistic quality through a continuous volumetric representation. However, the promise of high fidelity came with the constraints of slow training and rendering times, as the implicit nature of the representation necessitates numerous queries along every ray to volumetrically render a single pixel. Subsequent works sought to bridge this gap through improved encodings [1, 2] and accelerated structures with efficient data management [3, 4], yet these approaches often sacrificed quality for speed or navigated complex implementation-specific optimization hurdles.

A parallel line of research, however, sought to circumvent the high computational overhead of continuous volumetric rendering by leveraging explicit, point-based representations. Early efforts in differentiable point-based rendering showcased the potential to bypass traditional ray marching, achieving diversification in speed with a manageable compromise in visual quality [5, 6]. Drawing inspiration from these predecessors and the inherent expressiveness of radiance fields, 3D Gaussian Splatting (3DGS) [7] was introduced, presenting a paradigm shift in explicit scene representation. The core insight of 3DGS is not simply the use of points, but the introduction of an unstructured collection of anisotropic, differentiable 3D Gaussian primitives that directly serve as the scene’s building blocks. These primitives are composed of spatial positions, a full covariance matrix, opacity, and view-dependent appearance via spherical harmonics, all of which can be jointly optimized through differentiable rasterization. This formulation inherits the benefits of volume rendering for scene optimization—seamlessly handling occlusions and discontinuous geometry—but avoids the need for costly processing by only excavating in space that is actually present.

This first nature of the explicit Gaussian representation unlocks a new landscape of applications within real-time and interactive systems. The explicit formulation not only enables very fast rendering pipelines achieving high frame rates [8, 9] but also creates a direct correspondence between individual primitives and the underlying scene entities. Therefore, 3DGS naturally supports a level of editability and manipulability that is inherently difficult to achieve with continuous, implicit fields [10, 11, 12]. This makes it a fundamental bridge between neural representations and the traditional graphics pipeline. The infusion of the flexibility and expressiveness of deep learning with the directness of a traditional graphics primitive has rapidly positioned 3DGS as a foundational technology, giving rise to a diverse body of work targeting dynamic scenes [13, 14, 15], scene generation and editing [16, 17], as well as SLAM systems [18]. This body of work has moved beyond mere view synthesis and into dynamic scene representation and robotic navigation. A critical need has emerged for a systematic and comprehensive taxonomy.

This survey aims to address this need by providing a structured and systematic overview of this rapidly expanding corpus of works on 3DGS. Unlike earlier surveys, we propose a novel taxonomy that is organized around two primary axes: the **architectural advancements** of the 3DGS model itself (proceedings in optimization, rendering, and robustness) and the **application domain** which the 3DGS model is an area of development. Our survey investigates the latest innovations to give far reaching coverage of rendering economy [19, 20], robustness to sparse inputs [21], and the use of novel appearance models [22]. In parallel, we also provide a detailed exploration into applications spanning large-scale representations [9], physical simulation [23], radiance fields for dynamic human and head reconstruction [24], and more physically-informed

and generative methods [25] in order to place our analytical lens on the runtime. By creating the methodological core with its practical domain, in our analysis we identify not just current performance, but underlying trends and widely accepted future challenges. Our goal is to provide a roadmap for seasoned researchers and newcomers alike, offering a clear understanding of the historical context and a dedicated outlook on the open challenges and the potential trajectories that will shape the next generation of 3D Gaussian Splatting.

## 2 Foundations and Mathematical Framework

### 2.1 Mathematical Formulation of Gaussian Primitives

The 3D Gaussian Splatting framework represents a scene not as a continuous volumetric function or a discrete set of surface elements, but as an unstructured collection of anisotropic, differentiable primitives. Each primitive acts as a local, probabilistic representation of space with associated photometric and geometric attributes. This formulation is chosen to reconcile the competing demands of expressiveness, controllability, and computational efficiency, avoiding the topology constraints of meshes and the unstructured incompleteness of raw point clouds while preserving an explicit, editable structure [7, 26]. The explicit nature of this representation is a central departure from implicit neural field methods like NeRF [1, 5], enabling direct access and manipulation of scene parameters.

The density distribution of each primitive is the core of this representation. Rather than a simple point with a spatial extent represented by a radius, the Gaussian is mathematically defined as the probability density function of a multivariate Gaussian distribution in 3D space. Let a Gaussian be centered at its mean position vector,  $\mu \in \mathbb{R}^3$ . Its spatial influence is governed by a full covariance matrix,  $\Sigma \in \mathbb{R}^{3 \times 3}$ . The scalar density contributed to a point  $\mathbf{x}$  is given by: ““  $G(\mathbf{x}) = \exp(-(1/2)(\mathbf{x}-\mu)^T \Sigma^{-1}(\mathbf{x}-\mu))$  (1) ““ This formulation allows the primitive to assume ellipsoidal shapes that are arbitrarily oriented in 3D space. This anisotropy is a crucial differentiator from isotropic point-based systems, allowing for the compact representation of flat surfaces with extended or planar primitives as opposed to a higher count of spherical ones [20]. To ensure that the covariance is a valid positive semi-definite matrix during optimization, it is not stored in a raw matrix form but is parameterized using a scale vector  $\mathbf{s} \in \mathbb{R}^3$  and a rotation quaternion  $\mathbf{q}$ . This decomposition guarantees that gradient updates will necessarily maintain a legal, meaningful covariance matrix for the differential rendering pipeline [7, 27].

The contribution of each primitive to the final image is further modulated by a learned scalar,  $\alpha$ . This opacity value governs the level of transparency of the Gaussian, combining with the spatial density to allow for volumetric-like blending. Rather than treating the spatial influence and the opacity as separate, they combine within a rendering equation where a single scalar contribution from a primitive is determined by their product, allowing for the construction of capturing complex effects [28, 5]. This explicit control over opacity, which has a direct physical mapping, contrasts with the implicit, often poorly-understood density predicted by a multi-layer perceptron in standard neural network. The interplay of spatial extent and opacity is crucial for representing the concept of "surface" and for accurately handling partial occlusions and internal structures in the scene.

Finally, appearance is not stored as static RGB values. Instead, color is encoded as a function of the view direction, which is parameterized by a set of Spherical Harmonic (SH) coefficients. This is a computationally efficient basis for representing angularly varying radiance, capturing the appearance of how a surface changes with viewpoint. The rendering of color,  $\mathbf{c}$ , for view direction  $\mathbf{v}$  is formulated as: ““  $c(\mathbf{v}) = \sum_l \sum_m c_{lm} Y_{lm}(\mathbf{v})$  (2) ““ where  $Y_{lm}$  are the SH basis functions, and  $c_{lm}$  are learned coefficients. While a limited number of coefficients is capable of representing diffuse and low-frequency specular reflections, this SH basis often struggles to represent high-frequency specularities or anisotropic reflections [22]. Methods

such as anisotropic spherical Gaussians (ASG) provide an alternative, better suited to capture high-frequency appearance than SH terms, while keeping the representation efficient [22]. The challenge of representing complex surface appearances, especially for surface-aligned Gaussians, has motivated further exploration into alternative parameterizations, per-primitive texture maps. This decouples appearance from the geometric template, allowing modern primitives to capture scalable and high-resolution, complex textures [20].

While the original formulation has proven to be a strong starting point, its limitations have catalyzed a rich body of research. The choice of SH representation is far from universal, and investigations into alternate appearance representations—such as anisotropic spherical Gaussians [22], neural texture fields, and additional features—have sought to overcome the limitations of simple basis functions. The initial Gaussian mapping has targeted balance between geometry and appearance, allowing the fixed distribution to also govern the spatial and coordinate coordinates, as in the 2D Gaussian splatting [20]. This ongoing evolution of the primitive’s core mathematical structure, refined to achieve a balance between quality, flexibility, and memory efficiency, remains an active and central research direction in the field [26, 29].

## 2.2 The Tile-Based Differentiable Rasterization Pipeline

The tile-based differentiable rasterizer constitutes the computational engine of 3D Gaussian Splatting, transforming the unstructured collection of anisotropic primitives described in the previous section into discrete 2D imagery through a carefully orchestrated pipeline of projection, culling, sorting, and alpha compositing. Its design philosophy is fundamentally distinct from the ray-marching paradigm of neural radiance fields [1], opting instead for a forward-splatting approach that leverages modern GPU rasterization primitives to achieve unprecedented rendering throughput. This forward-splatting strategy is a direct consequence of the explicit, parameterized nature of the Gaussian primitives: each primitive’s closed-form mathematical definition enables a fully differentiable, hardware-accelerated mapping from continuous 3D space to the discrete 2D image plane, bypassing the costly per-ray sampling that defines implicit methods. The pipeline’s modular decomposition not only facilitates parallel execution but also defines the precise pathway through which gradients flow from the rendered image back to the primitive parameters—linking directly to the optimization and adaptive control mechanisms presented in the following subsection, which rely on the rasterizer’s output as their supervisory signal.

The initial computational step involves projecting the collection of 3D Gaussian primitives onto a 2D image plane. Since the projection of a 3D Gaussian is not exactly a 2D Gaussian under a full perspective transformation, the method employs a local affine approximation of the projection, leading to a closed-form solution for the 2D covariance matrix,  $\Sigma' = \mathbf{J}\mathbf{W}\Sigma\mathbf{W}^T\mathbf{J}^T$ , where  $\mathbf{W}$  is the view transformation matrix and  $\mathbf{J}$  is the Jacobian of the affine approximation. This 2D covariance is derived directly from the anisotropic 3D covariance parameterized by the scale vector and rotation quaternion discussed previously, ensuring that the projected footprint retains the ellipsoidal shape flexibility essential for capturing surfaces at arbitrary orientations. This analytical projection remains an approximation, stemming from the linearization of a fundamentally non-linear perspective transform. While the original formulation accepts this first-order linearization, subsequent work has investigated both the limitations of this error and ways to refine it, with [30] addressing sorting errors to improve near-field consistency. The accuracy of this 2D covariance is pivotal; it directly defines the spatial footprint and orientation of the splat on the screen, thereby governing the fine-grained shape and scale of resulting visual artifacts.

Following the projection phase, the rasterizer employs a culling and sorting strategy to manage the computational load inherent in processing millions of primitives. The screen is first divided into a grid of non-overlapping tiles, and each projected Gaussian is assigned to a tile based on its footprint’s intersection with the tile frustum. To maximize parallelism, per-tile visibility lists are constructed using highly efficient GPU kernels with atomic operations, a design

choice that avoids complex data structures and fully exploits modern graphics hardware. Within each tile, the Gaussian primitives are then sorted by view-space depth. The pivotal simplification here is the assumption of an invariant depth per primitive—a single depth value for the entire splat. This simplification is computationally efficient but is precisely what leads to "popping" artifacts at the seams of dynamic motion, a problem which [30] addresses. While the original model functions with a globally sorted list, exploring hierarchical or per-pixel depth sort to further enforce consistency remains an open, rich area for future advancement.

The final compositing stage involves alpha blending. For each pixel, the visible splats—obtained after tile selection and depth sorting—are traversed from front to back. The final color  $C$  is computed by integrating the color  $c_i$  of each splat, weighted by its opacity  $\alpha_i$ —the learned scalar discussed in the primitive representation—and the accumulated transmittance  $T$  of all preceding splats:  $C = \sum c_i \alpha_i T_i$ , where  $T_i = \prod_{j=1}^{i-1} (1 - \alpha_j)$ . This transmittance-weighted blending is a key differentiator, building upon the traditional alpha-compositing concepts formalized in [31] and refined within modern neural rendering pipelines [32]. Notably, this compositing stage is where the Spherical Harmonic coefficients—introduced earlier for view-dependent appearance—are evaluated with the view direction to produce each splat’s color  $c_i$ , thereby completing the mapping from the primitive’s photometric attributes to the final image. This final, fully differentiable step links the learnable parameters—positions, covariance, opacity, and SH coefficients—to the pixel value, enabling the gradient-based optimization framework of the next section. This stands in contrast to older, non-differentiable point-splatting systems, where such a direct gradient path was absent.

The true value of this pipeline lies in its pragmatic implementations; each stage, from an efficient custom CUDA kernel for forward rendering to a parallelized backward pass, is intricately co-designed with hardware capabilities. For example, the "single kernel" approach, as popularized by the original 3DGS pipeline [7], contrasts with the modular design of alternatives like gsplat, which adapt modular layers and tensor core-optimized operations. Further tuning, such as padded sorting and warping primitives, is crucial to mitigate memory-bound throughput bottlenecks. The arrangement of the data and the adjustment of tile size are major contributors to the empirical difference between achieving near-real-time vs. interactive frame rates, especially in large-scale scenes [9]. This design coalesces into a representation that is not a mathematically pure, continuous volume but an explicit, hardware-friendly particle system, capable of rendering high-resolution images in real-time [33], a feat that was previously unattainable with the complex layer of rendering costs associated with continuous coordinate-centric models [34]. This representation subsequently defines a fertile ground for the development of optimization and adaptive control strategies that operate directly in the parameter space, as will be discussed in the following sections.

## 2.3 Optimization, Loss Functions, and Adaptive Density Control

The optimization of 3D Gaussian Splatting (3DGS) is a delicate interplay between photometric fidelity, geometric plausibility, and topological adaptation of the primitive set. The canonical training objective combines a pixel-wise  $\mathcal{L}_1$  loss with a multi-scale structural dissimilarity term,  $\mathcal{L}_{D-SSIM}$ , to balance per-pixel accuracy against perceptual and high-frequency detail retention [7]. While this dual-loss formulation is effective for well-sampled scenes, it offers no explicit geometric constraint, which often leads to the formation of "floaters" or the misalignment of primitives with the physical surface—a critical weakness for downstream applications like meshing [20, 35]. To address this, a substantial body of work has introduced regularization terms. For instance, depth and normal priors can be leveraged to enforce consistency and smoothness on the Gaussian cloud, grounding the optimization in physical reality [35, 21, 36]. These geometric regularizers are not merely anecdotal additions; they fundamentally alter the optimization landscape by mitigating the ambiguous gradient signals propagated from photometric loss alone, thereby reducing the risk of local minima in under-constrained regions [37, 38]. The quality of the initialization is the

second pillar of this training process. The standard practice of seeding the optimization from a sparse Structure-from-Motion point cloud is both a strength and a critical vulnerability: it provides robust, semantically meaningful starting points for the initial Gaussians but becomes a significant challenge in texture-less regions or when the input data is severely degraded [39, 40]. This fragility has logically motivated research into COLMAP-free and random initialization strategies, where the reliance on pre-computed camera poses is removed, and the representation must be grown "from scratch," guided by the gradient signal and the spatio-temporal continuity of the input video stream [40]. While such methods improve robustness, their convergence is tied to careful scheduling and exposes the optimization to new kinds of failure modes.

At the heart of the topological adaptation is the adaptive density control (ADC) algorithm—the procedural engine that adjusts the population of Gaussians during optimization. This is not merely a heuristic step but an integral component of the training process, dynamically iterating between cloning and splitting primitives based on large view-space positional gradients [7]. However, this heuristic is known to be flawed. A primary failure mode is frequently caused by artifacts of this strategy, such as the failure to recover fine details due to large Gaussians being misclassified, producing blurry outputs [41]. This insight has catalyzed a new generation of densification strategies that are more principled for the task of density control; instead of relying on cumulative average gradients, more recent approaches propose a pixel-error-driven formulation [9] or pixel-aware gradient weighting [42]. These methods acknowledge that under-sampling in over-occluded or boundary regions yields a different scale of severity and that the generation of Gaussians must be driven by the reconstruction error per image, not just the mean geometric drift. Concurrently, the opacity handling in cloning is also being revisited to correct for the inherent bias present in the original strategy [43].

Looking forward, the trajectory of this theory is moving away from ad-hoc heuristics and toward a more loss-driven and probabilistic framework. The optimization of 3DGS is being reframed from a pure reconstruction problem to one of inverse rendering, where the goal is not just rendering images but recovering a quasi-materially-meaningful and stable representation [36, 20]. Future research will likely focus on creating theory for understanding how the ADC and the parameterization itself facilitate this, possibly leading to the generation of geometry-aware initializations that are less sensitive to the apparent critical reliance on SfM, thereby establishing 3DGS as a universal representation for the general world [44, 37].

## 2.4 Comparative Analysis with Alternative Scene Representations

The position of 3D Gaussian Splatting (3DGS) within the broader ecosystem of 3D scene representations is best understood not as a wholesale replacement of prior paradigms but as a strategic re-convergence of their respective strengths into a single, coherent framework. To appreciate this, one must consider the dominant alternatives—implicit neural radiance fields (NeRFs), explicit point clouds, and triangle meshes—as points on a design spectrum defined by axes of fidelity, efficiency, editability, and geometric robustness. 3DGS emerges as a novel hybrid that navigates these trade-offs with a precision that existing methods cannot match, framing the optimization dynamics discussed in the previous section as the process that unlocks this representational power.

The most fundamental comparison lies with neural radiance fields. NeRF-based methods [2] provide a continuous, volumetric scene representation by encoding geometry and appearance in a multi-layer perceptron. This in-built continuity affords them remarkable compactness and a smoothness prior that is highly effective for sparse or noisy input. However, this power comes at a practical cost: they typically require millions of network evaluations per rendered image, placing inference speeds far beyond the real-time threshold. 3DGS, with its explicit, discrete primitives, is the antithesis of this—the tile-based differentiable rasterizer transforms scene rendering into a memory-bound, parallel-friendly problem within a single kernel, achieving real-time performance. **Yet, this efficiency is intrinsically tied to the stochasticity and discrete nature of its**

**primitives.** NeRFs operate on a continuous distribution, yielding a spatial smoothing prior that natively handles sparse-view inputs. In contrast, the 3DGS optimization process—including the adaptive density control described earlier—more readily overfits to geometry observed from a limited number of views, leading to inaccuracies and "floating" artifacts. This is not merely a practical difference; it represents a fundamental shift from a smooth, globally-optimal estimator to a greedy, discrete data-fitting approach, necessitating the advanced regularization and initial-conditioning strategies explored in the preceding optimization section to achieve robustness comparable to NeRFs [45, 46].

The second dimension of analysis is with meshes, a representation for which 3DGS is frequently proposed as a replacement. Meshes are inherently topological and, for many real-world scenes, this explicit topology is a fatal flaw. Reconstructing a watertight manifold from a point cloud is a challenging, often ill-posed problem for scenes with vegetation, intricate structures, or non-manifold surfaces; discrete surfaces also typically require significant anti-aliasing for accurate rendering. Meshes are also the target of complex physically-based rendering (PBR) and painting pipelines. The crucial insight of 3DGS here is its effortlessness: it forgoes the need for a surface-level topological prior altogether, instead leveraging the statistical properties of a dense set of independent primitives (Gaussians) to define a "fuzzy", volumetric representation—a sum of Gaussians inherently different from a binary, step-function surface prediction [47]. This eschews the topological artifacts of meshes while preserving the editable flexibility of point-based graphics. The price, however, is the loss of an explicit "surface"—a property fundamental to geometry extraction, animation, and shape retrieval. Consequently, **significant research effort has been dedicated to extracting surfaces from the learned, volumetric nature of the Gaussian fields**, using various regularization and density control to compress the fields into a tight, surface-representing bound [37, 48]. This highlights a central tension: 3DGS excels at rendering *appearance* but is, by default, poorly suited to be a topological, watertight geometric representation—which remains the definitive strength of meshes.

To understand the mechanism behind this, we must place 3DGS alongside its most direct ancestors: point clouds. A conventional point cloud is a set of unordered, discrete samples with geometry (position) and perhaps a color. They are entirely non-differentiable for image synthesis. In this light, the 3D Gaussian primitive is a radical innovation. By associating a Gaussian distribution (covariance) with each point and introducing volumetric  $\alpha$ -blending, 3DGS transforms a sparse point cloud into a continuous, differentiable *scattering* function, elegantly solving the point proximity and occlusion problem that plagues standard point-based rendering. Moreover, spherical harmonics as a functional basis for view-dependence is a fundamentally more expressive appearance parameterization than the simple point colors of point clouds, and vastly more compact than the enormous MLPs of NeRF. The angular dependence granted by spherical harmonics fundamentally interplays with the shape of the Gaussian, mirroring the way particle scattering yields directionally-diverse radiance. From a statistical standpoint, 3DGS can be *interpreted* as a learned **Gaussian Mixture Model**, robust to noise and outliers [49]. However, it stands apart from canonical probabilistic models: the density (opacity) is never normalized, and its anisotropic covariances are not optimized as a probability function but solely as rendering parameters, yielding extreme efficiency and geometric flexibility at the cost of probabilistic interpretability.

Robustness to error and memory footprint further delineate these representations. Implicit representations, through their continuous prior, exhibit robustness against noise—but they can struggle with high-frequency details or geodetic drift. In stark contrast, **3DGS is highly pixel-data-responsive, which is a precision advantage but can become a robustness liability**: minor camera pose errors or matching noise will be directly embedded into the unconstrained optimization, leading to inconsistent renderings or "fuzzy" artifacts unless countered with geometric regularization or new adaptation mechanisms [50]. In terms of texture and footprint, a fully optimized 3DGS holds many primitives to encode high detail, incurring a **steeper memory**



**footprint** compared to a compressed neural network, though a significantly smaller cost than a high-poly mesh with texture maps. This has motivated significant investigation into pruning and minimizing the set of primitives, introducing strong capacity constraints to balance visual quality against an order-of-magnitude compression [45, 19]. These new optimization and compression strategies directly connect this representational analysis back to the implementation efficiencies discussed in the following section, where hardware-aware engineering aims to tame this intrinsic memory pressure.

In conclusion, the 3DGS framework transcends the capabilities of any single implicit, mesh, or point-based representation, but its unique status as a non-implicit, non-mesh, "hybrid {density + parametric} point of primitive" is simultaneously its major liability. The survey maps a clear trajectory forward: it is not a question of whether 3DGS is definitively "better" than NeRF, but rather how to mitigate its discrete fitting bias, increase robustness to sparse and noisy inputs, and impose a surface while retaining the speed. The future of the field, covered in the remainder of this survey, is dedicated to pushing the boundaries of this initially fragile form—integrating the semantic, geometric, and physical priors discussed in the optimization and implementation sections to solidify its status not just as a neural representation trick, but as a **foundational, robust, and editable 3D representation for the digital domain**.

## 2.5 Practical Implementation and Rendering Libraries

The practical realization of 3D Gaussian Splatting (3DGS) hinges on a sophisticated ecosystem of software libraries that translate the mathematical formulation of Section 2.1 into high-performance, differentiable GPU kernels. The foundational implementation by Kerbl et al. [7] established the canonical architecture, introducing a highly specialized, custom CUDA rasterizer that remains the performance baseline against which all subsequent implementations are measured. This reference implementation encapsulates the entire forward and backward passes into a single fused pipeline, interleaving the projection, sorting, and alpha-compositing stages to minimize memory bandwidth and kernel-launch overhead. The development landscape has since expanded into a modular software ecosystem, with libraries such as *gsplat* providing a more flexible, PyTorch-centric API. These libraries decompose the rasterizer into discrete, callable operators—positioning projection, covariance computation, and the core rasterization as separate modules [7, 26]. This decoupling is a critical practical foundation; it enables researchers to experiment with alternative appearance models, such as anisotropic spherical Gaussians [22] or per-primitive textures, without needing to re-engineer the underlying rendering engine, and it accelerates the field’s primary development pace by allowing rapid, modular experimentation.

The pursuit of efficiency has driven significant innovation in the design of high-performance rasterizers. A primary implementation concern is the management of the tile-based sorting algorithm. The original approach uses a single GPU kernel for sorting and global memory-intensive operations, which can become a bottleneck [7]. Contemporary libraries have adopted more sophisticated approaches, moving towards per-tile sorting and leveraging warp-level primitives to reduce scheduling overhead. Notably, FlashGS [51] introduces a tiled, software-managed data pipeline to maximize shared memory utilization and cache hit rates, directly addressing memory-access latency, which is often the primary performance limiter. Similarly, hierarchical approaches [9, 52] have been proposed to address artifacts from the constant-depth-per-primitive assumption, a trade-off discussed in Section 2.2. By representing the scene or sorting key as a hierarchical structure, these methods mitigate popping artifacts with a minimal speed penalty, a critical balancing act between algorithmic fidelity and implementation constraints.

Beyond the core scientific research, practical frameworks must support integration with broader neural network architectures. The differentiable nature of the rasterizer, governed by the transmittance-weighted blending and its gradients, allows for backward propagation into Gaussian parameters  $\mu$ ,  $\Sigma$ ,  $\alpha$ , and SH coefficients. This makes 3DGS a viable module for

many learning pipelines, from camera tracking in SLAM applications [53, 18] and robust pose estimation [54] to generative modeling and image synthesis [55, 56]. A major technical catalyst in this direction is the ability to provide efficient, differentiable operators that integrate with automatic differentiation frameworks like PyTorch3D [57], empowering hybrid learning where the Gaussian rasterizer can supervise or preserve the explicit scene module and other processing elements handle additional full or learned inputs.

Looking ahead, the field appears to be converging towards portable implementations and heterogeneous hardware support. While the original CUDA implementation is state-of-the-art, the push for near-real-time, high-fidelity rendering on a wide range of devices—from mobile SoCs to VR headsets—is driving research into cross-platform gigantorasterizers, including shader- and compute-based renderers (e.g., Vulkan, Metal) and lightweight implementations tailored for web and mobile deployment [58]. The primary pressure points are efficient memory access and bandwidth utilization, which have sparked research into on-the-fly decoders and compressed representations, as seen in LightGaussian [51] and HAC [59], where a deeper understanding of the rendering pipeline and specific scene statistics enables better deployment and memory footprints. The future of 3DGS implementation lies in closer on hardware-algorithm co-design, with a trend toward highly specialized and flexible libraries that can support anisotropic, 4D spatio-temporal, and even more advanced primitives.

### 3 Core Optimization, Refinement, and Robustness

#### 3.1 Enhanced Fidelity and Anti-Aliasing

The fidelity of 3D Gaussian Splatting (3DGS) is fundamentally challenged by the discrete and scale-sensitive nature of its primitives. The original model, trained at a fixed sampling rate, exhibits significant aliasing artifacts, including jagged edges, popping, and blurred textures, when the viewing distance or image resolution deviates from the training distribution [7]. This degradation arises because the projected size of a Gaussian on the image plane, which determines its footprint and contribution to a pixel, becomes critically misaligned with the pixel sampling rate. To combat this core limitation, a growing body of work has been dedicated to enforcing multi-scale consistency and mitigating high-frequency artifacts, addressing this at the spectral, geometric, and representational levels.

A foundational approach to mitigating aliasing involves directly controlling the spectral content of the Gaussian primitives during optimization. As pointed out by [41], the adaptive density control strategy often suffers from a "gradient collision" problem, preventing large Gaussians in over-reconstructed regions from splitting. This can lead to blurry renders and an inability to create the necessary higher-frequency primitives to match the training sampling rate. To address this, a core direction is to regulate geometry by incorporating an analysis of the primitive’s spectral footprint (i.e., its covariance matrix) during densification. By analyzing these intrinsic geometric properties, methods can guide the splitting of Gaussians more effectively, preventing overfitting to high-frequency details that are not view-consistent and avoiding the creation of needle-like artifacts when observed at higher frequencies than the training data could accurately capture. Concurrently, a different set of works identifies the core of the problem not in training, but in the test-time rendering pipeline itself. These methods propose to filter the spectral content of 3D Gaussians by applying scale-adaptive smoothing and filtering operations, ensuring that the rendering process does not display frequencies higher than the training sampling rate can faithfully represent, effectively acting as an anti-aliasing mechanism based on the physical frequency of the primitives.

A particularly impactful and computationally efficient avenue is the development of training-free, test-time filtering methods. These techniques act as a post-hoc plugin to the rendering pipeline, adapting the effective scale of the projected Gaussians to match the target view’s

sampling density. This is often achieved by estimating the footprint of the Gaussian in pixel space and, if it does not cover a sufficient area to be accurately represented, the method adjusts its effective coverage to prevent undersampling. By designing this to be scale-adaptive, it effectively reduces aliasing and jaggedness during zoom-in and zoom-out without altering the original training procedure [8]. This offers a foundational and training-free alternative that can be seamlessly integrated into existing pipelines, ensuring a significant reduction in aliasing artifacts and jaggedness at minimal computational cost. These methods have proven to be a robust solution to the inconsistency between the native resolution of the training dataset and the desired test-time view for real-time applications.

Addressing the problem holistically, multi-scale and mipmap-based approaches introduce scale-consistent guidance directly into the training process. Rather than optimizing a single model, these frameworks co-train Gaussian primitives across several scales, for instance by rendering the scene at multiple resolutions and applying a scale-consistency loss to the output. This multi-scale supervision signal effectively acts as a strong regularizer, enabling the 3D Gaussians to self-adjust their color, shape, size, and position across different levels of detail [52]. Furthermore, advanced approaches incorporate a hierarchical, LOD-structured representation of the scene, where multi-resolution anchor points select appropriate levels of detail to ensure smooth transitions and maintain high-fidelity rendering at any scale. This approach distinguishes itself from simple LOD by enabling a continuous representation of the scene, thereby ensuring fidelity across all zoom levels without discrete jumps in quality [52]. This ensures fidelity across all zoom levels without discrete jumps in quality.

A major performance bottleneck is that the original model has an unbounded, open set of primitives that struggles to quickly adapt to new views, leading to an excessive number of Gaussians to fill in missing details. This is particularly problematic for dynamic scenes with large variations in depth [15], where slight inaccuracies in geometry directly manifest as flickering or ghosting artifacts. This can be mitigated using test-time adaptive density control, like splitting the points and, importantly, using the additional information from the input to guide refinement. To couple the distinct and dynamic nature, a method that minimizes the directional and temporal error is necessary. By assuming and enforcing local-rigidity, we can constrain the problem, but the under-constrained regions require flexible strategies for distractor handling of the sparse and non-Lambertian surfaces. Therefore, solutions are not uni-dimensional, but require a hierarchy of processing strategies.

In conclusion, the field shows a clear evolution of approaches from the base representation to the scale-dependent nature of splatting. The shift from simple geometric scaling to more sophisticated strategies based on spectral analysis, probabilistic filtering, and multi-scale supervision illustrates how the core strength of the explicit Gaussian representation—its adaptability—is equally critical for its final visual fidelity. The field is also finding a natural convergence with dynamic scene methods [14], where the manifold of "physical time" also needs to be sampled and anti-aliased upon to avoid jumps over time. The most promising future direction likely lies in the unified approach, which harmonizes the explicit view-dependent, resolution-aware nature of the Gaussian primitives with the latest advances in regularization. It asserts that enhanced fidelity is not the final goal but a necessary prerequisite for achieving true effectiveness. We cannot separate the objective from the scale; instead, the aspects of "quality" and "scale" must be synthesized.

### 3.2 Geometric Refinement and Surface Extraction

The original 3D Gaussian Splatting (3DGS) formulation, while achieving remarkable rendering fidelity, produces a volumetric "fluffy" cloud of anisotropic ellipsoids that are inherently misaligned with the underlying physical surfaces. This fundamental limitation stems from the unconstrained nature of the 3D Gaussians, which lack a defined surface orientation and are optimized solely for photometric consistency. Consequently, extracting a clean, renderable mesh asset via standard

algorithms like Marching Cubes or Poisson reconstruction remains a formidable challenge, as the geometry is riddled with floaters, lacks topological coherence, and exhibits high-frequency normal noise. This issue is further compounded by the scale-dependent artifacts discussed in the previous section: even when a geometrically accurate optimization is achieved, the multi-scale inconsistencies and spectral noise introduced during training directly propagate into the extracted surface, obscuring fine structure and creating spurious high-frequency geometry. To bridge the gap between point-based radiance fields and explicit surface representations, a substantial body of research has focused on geometric refinement and surface extraction, aiming to regularize the Gaussian primitives into a form amenable to surface modeling—building upon the same principles of spectral and structural control that underpin anti-aliasing and scale-consistent rendering.

A primary line of investigation seeks to reshape the primitive itself, advocating for surface-aligned kernels. In contrast to the original 3D ellipsoids, **2D Gaussian Splatting (2DGS)** [20] collapses the volume into a set of 2D oriented planar Gaussian disks. These disks are inherently view-consistent and lie flat on the surface. By employing perspective-accurate ray-splat intersection and rasterization, alongside depth distortion and normal consistency terms, 2DGS provides a new paradigm where the geometry is encoded into a dense set of flat primitives, facilitating direct surface extraction and achieving high-quality normal maps. This approach is mirrored in **Gaussian Opacity Fields (GOF)** [60], which uses ray-tracing-based volume rendering and derives a smooth and compact implicit field directly from the Gaussian parameters. GOF’s contribution lies in its efficient geometry extraction method using marching tetrahedra, a process that is adaptive to the scene’s complexity. A similar need to decouple motion from appearance is seen in avatar reconstruction, where **SplattingAvatar** [24] embeds the Gaussians onto a triangle mesh, while **GaMeS** [61] parameterizes Gaussian components by mesh vertices, allowing for real-time animation and modification of 3D content in a mesh-like fashion. In a distinct approach, **GeoGaussian** [62] targets the degradation of geometry in non-textured regions. It introduces a geometry-aware initialization and densification pipeline that creates thin Gaussians aligned with surfaces, demonstrating that maintaining the physical integrity of boundaries is essential for improving novel-view synthesis.

Complementing these primitive-centric modifications is a wave of research focused on regularizing the standard 3D Gaussian kernels to estimate robust surface normals, address anisotropic issues, and eliminate redundant primitives. A major challenge in this domain is the accurate estimation of normals from the discrete, unstructured nature of 3D Gaussians, as it is the linchpin for applying physical shading functions and surface constraints. For instance, **GaussianShader** [63] proposes a simplified shading function coupled with a novel normal estimation framework based on the shortest axis direction, ensuring consistency between normals and the geometry. **2DGS** [20] likewise introduces normal consistency and depth distortion losses to enforce smooth local surfaces. To further bridge the gap between a renderable radiance field and a usable mesh, the necessity of regularization is paramount. Many works propose combining the speed of 3DGS with the geometric stability of implicit signed distance functions (SDF). **3DGSR** [37] incorporates a differentiable SDF-to-opacity transformation into the Gaussian splatting pipeline, uniting the two representations. This allows for the direct optimization of the SDF and the Gaussians in a cohesive manner, ensuring that surfaces outside the Gaussian sampling range are effectively eliminated, thereby also complementing the above geometric constraints for sparse-view robustness. Yet, as suggested during the earlier discussion around **GaussianAvatar** [64], malformed observations often struggle with the inherent discontinuity of the point-splat, reinforcing the need for priors akin to those described previously.

Recent advancements have also centered on improving the surface extraction process itself, moving beyond raw geometry to rigorous spectral and density control—echoing the anti-aliasing strategies discussed earlier. The issue of erratic "floater" formation and over-reconstruction often arises from the basis of geometric loss functions and optimization workflow of the original

3DGS. To address these issues at a more internal chemical level, **AbsGS** [41] introduces the phenomenon of "gradient collision," demonstrating that this flaw leads to blurry renderings in highly detailed regions. By changing the densification criterion from the average unity-space dynamical gradient to a homodirectional one, AbsGS provides a tool that more efficiently identifies and splits large, ambiguous Gaussians, promoting cleaner geometry and inadvertently making the model more robust to gradient-scale asymmetries. Similarly, **Pixel-GS** [42] corrects the gradient by weighting it with the number of pixels covered by a Gaussian in each specific view. This prevents large splats that are only observed at their boundaries from becoming stuck in a state of overestimation, acting as a filter to retain fine coverage without sacrificing data. Following a more structured approach to content management, **Mini-Splatting** [7] partitions (i.e., blur split) and reinitializes depth information to reorganize the spatial distribution of point sets, leading to more efficient use of the delimited number of primitives to achieve cleaner results and a lower memory footprint.

Looking ahead, the field is making a decisive shift from photorealistic approximations to tangible, structured virtual assets. The trajectory points toward hybrid systems that can act as a bridge between the fidelity of 3DGS and the topological flexibility of meshes, echoing the shift from pixel-perfect radiance fields to physically grounded primitives. Future research will likely focus on more sophisticated relationships between a coarse SDF/Mesh and the Gaussians, enforcing light transport on meshes in conjunction with high-fidelity rendering. There is also an increasing convergence between appearance and physics, with high-posed tasks like **PhysGaussian** [33] demonstrating the necessity of physically grounded deformation to maintain a plausible and continuous geometry even under dynamic stress. This, in turn, necessitates a tighter integration between the estimated surface and the intra-parameter correlations between the Gaussian primitives to ensure that physical deformation is represented with the same fidelity as the novel-view synthesis—yet the foundation of this fidelity remains anchored in the scale-consistent, aliasing-free representations we have refined so far.

### 3.3 Robustness to Sparse and Degraded Input Data

The canonical 3D Gaussian Splatting pipeline presupposes dense, high-quality multi-view captures with reliable Structure-from-Motion (SfM) initialization, a precondition that is frequently violated in real-world deployments. When the number of input images is severely limited, or when captures are degraded by motion blur, defocus, or sensor noise, the optimization becomes severely under-constrained. The inherent flexibility of the explicit Gaussian representation, which is a strength in well-conditioned scenarios, transforms into a liability: the model overfits to the sparse photometric observations and produces geometry that is replete with floaters, high-frequency artifacts, and semantically incorrect volumetric "fluff" rather than a coherent surface. Consequently, a substantial body of recent research has moved away from relying solely on 2D photometric loss and instead advocates for integrating external priors to impose strong geometrically plausible constraints, thereby regularizing the ill-posed optimization landscape.

A prominent and highly effective strategy is the use of depth-guided regularization derived from large-scale pre-trained monocular depth estimation networks. The core premise is that although the monocular depth predictions are only up to an unknown scale and shift, their relative geometric structure provides a powerful, semantically consistent prior for the scene. Frameworks such as **DNGaussian** [21] devise a "Hard and Soft" depth regularization that aligns the rendered depth with coarse monocular predictions to arrest the proliferation of inaccurate geometry. A critical innovation in this line is the **Global-Local Depth Normalization** strategy which aligns the normalized global scale and then sharpens the model's focus on small, localized depth variations, creating a sizable performance difference when compared to un-normalized constraints, successfully balancing the need for global consistency with the desire for fine-grained local detail. However, this alignment is only reliable when the monocular and SfM estimates are on the same relative scale. Therefore, other works such as **DN-Splatter**

[35] integrate both depth cues and local normal smoothness constraints between neighboring Gaussians, demonstrating that enforcing local surface flatness via normal consistency helps indoor scenes replete with large textureless planes, facilitating direct mesh extraction. Despite their effectiveness, these depth-prior strategies remain sensitive to outliers and can suffer from the inherent depth ambiguity of monocular estimation, especially in scenes with non-Lambertian surfaces or translucent objects.

A second major axis of research tackles the brittleness of SfM initialization itself, which often fails catastrophically in sparse or textureless regions. Instead of relying on the often imprecise SfM point cloud, **geometry-aware initialization** leverages learning-based multi-view stereo (MVS) to generate a dense and more reliable prior. For instance, methods exemplified by **GaussianPro** [39] adopt a progressive propagation strategy. This approach drastically departs from the naive "split and clone" heuristic, instead using the existing reconstructed geometry as a scaffolding framework and employing patch-matching techniques to actively generate new Gaussians with accurate positions and orientation in vacant regions. By using the MVS-prior as an optimal initial scaffolding, it smoothly grows Gaussian primitives onto texture-free surfaces, preventing the flat and concave failures modes common to sparse-baseline setups. This indicates a paradigm shift toward active, geometry-guided propagation, where the algorithm uses depth cues rather than relying purely on photometric gradients. A recent variant explores the viability of training fully random initialization frameworks via **COLMAP-Free 3D Gaussian Splatting** (COLMAP-Free 3DGS) [40], which explicitly rejects the necessity of pose estimation. By processing input frames sequentially and progressively growing the Gaussian set based on their continuity, the method estimates poses on-the-fly. While this addresses the SfM fragility, it equally warrants additional geometric constraints to prevent drift over time.

The third class includes solutions designed for **blur-constant treatment**. The key challenge is to deblur the scene instead of learning the blur as a baked-in artifact. **Deblurring 3D Gaussian Splatting** [65] introduces a compact neural network (MLP) colocated with the Gaussians to model the blur process through the per-Gaussian covariance modulation, enabling the recovery of a sharp, canonical representation from blurry training images. This is a natural and elegant adaptation: reasoning about 2D blur footprints and the 3D Gaussian per-primitive geometry, the renderer can retrieve a blurred image while the optimizer works in sharp space, transforming the blur into physically modeled parameters to repair the original details. This strategy ensures that the final model remains accurate and competitive to the blur-free case while keeping real-time rendering.

Beneath the comprehension of these explicit geometric and photometric priors, a more recent strategy is emerging: **the use of generative priors to serve as semantic guidance**. By leveraging the production-ready generation capability of large pre-trained 2D diffusion models, it becomes possible to effectively regularize newly rendered views from unobserved directions, significantly constraining under-constrained spaces. This is more relevant in dynamic captures, where the task is not simply novel view synthesis but also smooth, temporally coherent view interpolation. Simple geometric monocular priors fall short in degenerate or nearly degenerate situations such as a near-static camera or extreme motion. As seen in **Gaussian Splatting LK** [66], dynamic 3DGS variants can use the native warp fields to derive a velocity field analytically, and integrating it over time yields *scene-flow* constraints. This acts as non-trivial, auxiliary regularizations. The derived Lucas-Kanade style analytical regularization enables the method to outperform existing dynamic Gaussian Splatting frameworks even under minimal camera movement and sparse-time inputs, directly addressing the dynamic sparse-view limitation.

Synthesizing the findings, the trajectory is a conscious move away from generalization solely under the umbrella of pure photometric learning. Most sophisticated methods have shifted toward a form of "hybridization": incorporating belief depth, monocular depth, surface normals, and diffusion-based diffusion models to combat the natural information lack in sparse and degraded capture regimes. A critical trend is moving from the "vanilla" 3DGS objective into a

purely "multi-modal prior, robust optimization problem." However, a significant chasm remains in their theoretical underpinnings and theoretical uncertainty: existing depth priors do not always account for the geometry uncertainty or scene occlusions directly; there are no principled ways to determine whether global-depth regularization will harm appearance under strong photometric conflict. The prominent future evolution will be to develop uncertainty-aware priors that measure confidence in the external source and adaptively reweight its constraint online (e.g., reliability fusion for monocular depth/normal, or masking blurry frames). Additionally, with the acceptance of unfraigned/random basis methods, a fertile untouched ground remains in architecting smooth differentiable tracking to serve as robust SfM replacement, voluntarily encoded via a multi-layer, geometry-aware robust optimization plan. This model casts a perspective: robustness is not derived from regularization alone when only external priors are scarce, but from the orchestration of a suite of complementary constraints, balancing in the task with real data.

### 3.4 Handling "In-the-Wild" and Unconstrained Photo Collections

The canonical 3D Gaussian Splatting (3DGS) formulation assumes a static, radiometrically consistent world, where each point in space exhibits a fixed radiance that is solely a function of viewing direction. This assumption is fundamentally violated by "in-the-wild" photo collections, such as those harvested from online databases, which aggregate images captured by numerous users under drastically varying illumination, with diverse camera post-processing pipelines, and populated by transient occluders like pedestrians or vehicles. This challenge is orthogonal yet complementary to the geometric and photometric robustness issues discussed in the previous section: whereas sparse or degraded inputs suffer from an *information deficit*, unconstrained captures suffer from *information conflicts*—the same surface is observed under inconsistent appearance and with foreign, ephemeral content, jointly obscuring the true static geometry and reflectance. Reconstructing robust and high-quality scenes from such unconstrained data necessitates a paradigm shift from modeling a single static appearance to explicitly disentangling the persistent scene structure and reflectance from ephemeral, per-image variations. The primary research thrusts can be categorized into four complementary strategies: appearance-conditioned modeling, transient-object handling, semantics-guided decomposition, and uncertainty-aware fusion.

A prominent approach to handle appearance variation is to condition the view-dependent appearance of Gaussians on a per-image latent code. Drawing inspiration from the computer graphics concept of a "tone-mapper," methods employ a per-image appearance embedding or a residual-based module that modifies the base spherical harmonic coefficients, effectively adapting each Gaussian to the specific lighting and white-balance of an input image [67]. This latent modulation allows the optimization to disentangle the canonical scene albedo from the photometric transformations of each capture, yielding a more consistent aggregate synthesis and enabling appearance transfer between views. However, the representational capacity of a single residual vector can be a limiting factor for highly complex or *non-parametric* changes in illumination, such as dynamic shadows or specular highlight changes. Therefore, more granular approaches are required to decouple appearance on a per-primitive basis, similar to how depth and normal priors are selectively and locally injected into geometry optimization in the previous section.

Another critical component in handling the "in-the-wild" setting is the isolation of transient occluders and distractors, which otherwise corrupt the optimized geometry of the static scene. Analogous to the concept of distractor handling in classical photogrammetry and recent implicit methods [68], research has integrated "mask predictors" and residual feature maps directly into the Gaussian optimization process [69]. These components are designed to learn and identify per-image inconsistencies. By predicting a mask or a weighted transiency map, the optimizer can effectively down-weight or ignore the gradients originating from transient pedestrians or vehicles, preventing them from being fused into the canonical and geometrically clean scene representation.

Rather than fully discarding potentially valuable pixels—which is a known limitation in strong sparse-view regularization—some strategies aim for a more confident or consistency-driven fusion by combining information based on geometric and color reliability across observations, creating a form of robust "less-reliable observation aggregation" that parallels the uncertainty weighting discussed in depth-sensor fusion [41].

Semantic and structural priors are also increasingly leveraged to enforce consistency under variable conditions. By integrating foundation model features extracted from 2D models like SAM or CLIP into the 3D Gaussian field [69], methods facilitate semantic correspondences across images captured under vastly different times of the day or seasons. These correspondences act as robust anchors for optimizing geometry and appearance, mitigating the ambiguity introduced by the changing appearance and serving a role analogous to the MVS or depth anchors in sparse-view methods. Furthermore, architectural advances are increasingly incorporating per-Gaussian uncertainty to orchestrate which observations contribute most to the canonical model, creating a form of "confidence-aware rendering" that is directly extensible to fusion with multi-modal or multimodal depth signals.

Looking forward, the field lies at the intersection of generative and robust statistical models. Combining recent diffusion-based models as generation priors for occluded or unseen regions holds great potential to not just hallucinate but to "complete" a scene in a physically and semantically consistent way, significantly improving performance on unconstrained collections. Future research will likely pursue architectures that dynamically disentangle the latent appearance code into physically meaningful properties, such as scalar illumination intensity, correlated color temperature, and a static camera response function, thereby facilitating higher-level appearance editing and physically grounded relighting. In closing, the challenge of in-the-wild 3D reconstruction is thus not only a test of geometric fidelity but also a model’s ability to dynamically aggregate, weigh, and disentangle a multi-attribute world observation—an ability that will underpin the integration of dense, heterogeneous sensor inputs discussed in the next part of this survey.

### 3.5 Integration with External Sensors and Multi-Modal Depth Priors

The maturation of 3D Gaussian Splatting (3DGS) from a purely image-driven reconstruction framework into a versatile multi-modal scene representation paradigm hinges on its capacity to integrate external geometric cues. While photometric loss functions provide rich appearance supervision, they are fundamentally ill-posed for recovering accurate geometry in texture-less regions, under sparse view configurations, or in the presence of specularities. The integration of depth priors derived from active sensors, such as LiDAR or RGB-D cameras, and passive self-supervised monocular estimators, directly addresses this liability by funnelling explicit 3D constraints into the optimization.

A primary strategy couples 3DGS with established SLAM front-ends to jointly estimate camera pose and the Gaussian map both in the tracking and mapping iterations [53, 18]. To enhance the robustness of dense tracking, uncertainty-aware optimization frameworks have been proposed that estimate and discard unstable Gaussian primitives to favor the valuable ones for optimization without the stability tests, achieving tracking speeds up to 15 Hz and superior accuracy and mapping performance [53, 18]. However, the stability test in these integrated models is often limited, and their extension to laser-fused global frameworks remains an open frontier for large-scale simultaneity.

For the scenarios where the camera-SLAM baselines are restrictive or degenerate, the fusion of lower-frequency sensors like sound navigation and ranging is critical. In restricted underwater settings, the direct coupling of range and bearing data from sonar compensates for the deficits of sparse or missing viewpoints [53]. This synergistic fusion can directly support the reconstruction of geometrically consistent digital twins where visual features are scarce, underscoring the need for sensor-specific fusion rather than generic depth encodings.



Another powerful and accessible alternative to dedicated hardware is using learned geometric priors to regularize the 3DGS volume [35]. These losses enforce strong geometric constraints from off-shelf foundation models and are used in few-shot reconstruction pipelines [70]. Monocular priors and Spherical Harmonics activation are crucial as some high-dynamic-range specular scenes are extremely sensitive to depth map errors [22]. Its boundary coupling function robustly discards occlusions and latitude-aware checkpoints define a more general and robust deep regularization that is unsolved. A key challenge is to lock the monocular geometry and better encode the uncertainty and multi-view-instabilities into these priors, both of which are currently heavy heuristic and unsupported.

Generative priors are further funnelled into the optimization to address extreme sparsity [39]. A coarse-to-fine scheme can refine normals and patch-matching priors, mitigating over-smoothness and floaters in the absence of ground-truth geometry [70, 36]. The directional consistency in particular stabilizes the geometry-aware training and remains a fast remedy for the intrinsic high-frequency overfitting of Gaussian primitives.

Despite successes, the integration events between global depth sensors (LiDAR) and local monocular priors reveal a critical trade-off: the hard labels from LiDAR improve large-scale accuracy but do not close the data gap between high-frequency details and deformable objects. A new frontier is the decomposable scene placement of fusion of photometry + dense geometry maintained practically with spatially-varying fusion, incorporating a densification gate with pixel-adaptive weights and hierarchical continuous geometric context to lead to the trustful geometric priors, achieving both precision and cohesion. The systematic aggregation of these multi-modal cameras is the only sustainable path to truly scale 3DGS from as an offline optical rendering architecture to a complete, sampled, continuous geometrical representation.

## 4 Efficiency, Compression, and Scalability

### 4.1 Hardware-Accelerated Rendering and Rasterization Optimization

The relentless pursuit of interactive frame rates for 3D Gaussian Splatting (3DGS) has positioned the tile-based differentiable rasterizer as both the enabling technology and the primary computational bottleneck of the paradigm. While the foundational work of Kerbl et al. [7] established a highly efficient GPU pipeline, its viability across diverse consumer, mobile, and embedded platforms hinges on continuous, hardware-aware algorithmic innovation. This subsection dissects the principal axes of optimization, moving from kernel-level engineering to high-level scene structure exploitation, to achieve high-fidelity rendering beyond the capabilities of the initial implementation.

A primary focus of recent research is the meticulous refinement of the GPU execution path. The original rasterizer’s per-pixel complexity, stemming from sorting and blending millions of anisotropic primitives, presents significant opportunities for improved memory access patterns and compute utilization. To address these, the `gsplat` library has emerged as a modular and highly optimized CUDA backend, introducing warp-level primitives and more efficient parallel scan operations that decouple the forward and backward passes for enhanced flexibility and reduced memory overhead. Complementing this, FlashGS [71] directly targets memory-bound bottlenecks by implementing a vectorized, tile-based pipeline that improves cache locality and atomic operation efficiency through a "flash" scheduling mechanism. Empirical results from FlashGS demonstrate substantial FPS improvements—often exceeding 4×—across a variety of datasets by minimizing the latency inherent in memory-heavy read-write operations. This low-level optimization ensures that the theoretical parallel capacity of modern GPUs is more effectively translated into hands-on rendering speed.

Beyond raw kernel efficiency, eliminating redundant computation through sophisticated culling strategies is vital, particularly for large-scale or sparsely populated scenes. The naive

pipeline processes all Gaussians within a tile’s bounding box, failing to account for empty space or heavily occluded regions. Hierarchical culling mechanisms, such as occupancy grids and spatial data structures, directly address this inefficiency. Specifically, methods inspired by voxel-based acceleration, like those used in Octree-GS [52], and hierarchical culling approaches in RadSplat [8], facilitate the rapid identification and skipping of inactive or low-contribution Gaussians before the expensive per-pixel blending stage. By eliminating irrelevant work from the rendering equation, these effective pruning strategies reduce the sort list size and the computational load on the rasterization cores. This synergy between spatial indexing and the rasterizer advances scaling 3DGS to unbounded environments without compromising frame rate.

The final area of optimization is the design of the pipeline itself, specifically the formulations for sorting and blending. The original method relies on a global per-pixel sort, a costly operation that scales super-linearly with the number of primitives. Hierarchical rasterization schemes, such as those proposed in StopThePop [30], challenge this premise by introducing a hierarchical data structure that assigns a weight to the sort for improved efficiency. This design, while adding a mere 4% overhead in its baseline mode, is able to enforce view consistency when per-pixel depth is required, and by enabling a reduction in Gaussian count, it can ultimately achieve a  $1.6\times$  speedup over the original algorithm. This is complemented by algorithmic refinements in data scheduling and visibility-aware rendering that aim to make the algorithm bandwidth-bound rather than latency-bound [7]. Such holistic improvements ensure that the advanced high quality blending and level-of-detail use cases do not come at the cost of performance.

Looking towards the future, we are clearly at the frontier of fusing these optimization levels with the rapid development in AI compute. The emergence of concepts like tensor cores, and the desire to move beyond cuDA into operation which can be compiled for different platforms, suggests that more marginal returns could be achieved by leveraging level-of-detail structures and hierarchical methods [9]. The ultimate goal is an increasingly block-wise adaptive rasterizer that can reallocate compute per scene and per view, making high-fidelity, real-time 3DGS a ubiquitous reality across the digital ecosystem.

## 4.2 Model Compression, Pruning, and Storage Reduction

The inherent scalability of 3D Gaussian Splatting (3DGS) is bounded by its substantial memory footprint; a single optimized scene can require hundreds of megabytes due to millions of anisotropic primitives and high-order spherical harmonic coefficients. While the preceding discussion established the algorithmic foundations of the tile-based rasterizer, the practical deployment of 3DGS across consumer, mobile, and embedded platforms—particularly when paired with the real-time optimization strategies to be examined in the following subsection—hinges critically on solving this storage bottleneck. This subsection categorizes the primary strategies for compression and storage reduction, which are essential for enabling interactive frame rates and feasible memory usage before the rendering acceleration techniques can be meaningfully applied.

**Sparsity and Importance-Based Primitive Pruning.** A direct strategy for reducing memory consumption is decreasing the primitive count. Early approaches proposed resolution-aware pruning, which selects Gaussians based on their image-space footprint to eliminate those that are unnecessarily small or redundant, achieving roughly a  $2\times$  reduction in primitives [72]. More principled methods construct a graph on Gaussian primitives to perform spectral down-sampling, pruning nodes via a high-pass graph filter while a lightweight neural head compensates for the lost high-frequency signals [73]. Similarly, the method’s opacity and cache complexity have been used to prune prone-to-vacancy Gaussians [73]. Sample-based simplification, such as those used in Mini-Splatting, reframes density control as a sampling problem to achieve a constrained Gaussian count while maintaining rendering quality [19]. A comprehensive analysis from Reducing the Memory Footprint of 3D Gaussian Splatting further highlights that half-float precision for attributes, alongside pruning, leads to significant memory savings, achieving about

a  $27\times$  reduction for tested scenes [72]. In a similar vein, analytical approaches that introduce hierarchical sorting also prune half the Gaussians [30] during preprocessing, thereby aligning the reduced primitive count with faster rasterization in the rendering pipeline discussed later.

**Quantization and Efficient Coding.** Quantizing the high-precision floating-point values into lower-bit codes is a complementary, yet powerful, approach. A codebook-based vector quantization (VQ) method [72] learns compact codebooks for geometric attributes (scale and rotation), achieving a further storage reduction. This was more explicitly formalized in Compact 3D Gaussian (C3DGS), which combines VQ with learnable masks and grid-based alternative fields, reporting a consistent  $25\times$  storage reduction against vanilla 3DGS [27]. A key challenge for VQ is the non-uniform distribution of attributes across a scene. Recent methods [27] proposal to project the 3D Gaussians onto a self-organizing map, then quantize the co-located neural features, using quantized coordinates to enable attribute spaces to be more uniformly distributed and learned more effectively. These compact attribute representations necessitate that the subsequent rendering phase can efficiently decompose and access the decoded data; just as the previous subsection’s pruning and LOD techniques removed the overhead of accessing empty regions, the algorithm-level coordination for quantized data—such as alignment with the tile-based data structures discussed in the following subsection—becomes essential for translating storage savings into actual bandwidth reduction.

**Neural and Structured Compensation.** An alternative to pruning and quantization is to remove redundancy in the parameters themselves by using compact, spatially coherent representations. The encoding of spherical harmonics (SH) coefficients can be re-parameterized into a grid-based neural field [27], reducing the per-Gaussian storage from a high-dimensional vector to a low-dimensional feature lookup, with a small MLP to decode the color. Additionally, graph-based approaches [73] deploy a post-pruning neural compensation module that learns to mix the splatted features, which fills the un-pruned areas while lowering the parameter count. Such structured designs are particularly synergistic with the hierarchical and multi-resolution anchoring strategies discussed in the next subsection, because they create a compact intermediate value and level-of-detail mechanisms that are both cache-friendly and directly usable by the rasterizer’s culling stage.

The field exhibits several key trends. The most promising line of research lies in the combination of compression strategies into a holistic pipeline and algorithm-level co-design. For example, the combined use of pruning, quantization, and neural compensation in [27, 72] demonstrates that these methods are often complementary, achieving multiplicative reductions that surpass any single technique. A second trend is the shift from heuristic, hand-crafted schemes to learned, data-driven ones, as seen in the spectral graph filtering [73] and neural compensation networks [73]. Looking ahead, future research is likely to emerge in two important areas. The first is the extension of these compression schemes to dynamic or 4D Gaussian representations, where temporal coherence presents new opportunities for removing both spatial and temporal redundancy. The second is the establishment of a rate-distortion theory for Gaussian splatting, which would allow for a principled, task-agnostic approach to optimizing the bitrate (the number of Gaussians, the quantization level, or the neural field size) against the distortion (perceptual and pixel quality) [27]. More recently, relation to storage and bandwidth constraints when combined with the LoD culling and hierarchical schemes discussed in the following subsection—where reducing the number of active, full-precision primitives inside the renderer’s view frustum reduces both the byte footprint and the per-frame rasterization cost—highlights a broader co-design trend: compression techniques that treat rendering memory and compute as interchangeable constraints [13] and that prioritize encoding efficiency before hardware-level execution will be most effective. This co-design between what is stored (compression), what is rendered (LoD selection), and how it is rasterized (GPU-scheduling) is the key to realizing the full performance envelope of 3DGS on real-time systems.

### 4.3 Level-of-Detail and Hierarchical Structures for Large-Scale Scenes

The escalating scale of captured scenes, from room-scale environments to unbounded city models comprising tens of thousands of images, presents a fundamental challenge to the resource-intensive nature of vanilla 3D Gaussian Splatting. The naive approach of globally optimizing a monolithic set of Gaussian primitives becomes computationally intractable and memory-prohibitive as scene complexity grows. Hierarchical and level-of-detail (LoD) structures have emerged as the principal solution paradigm to address these scalability bottlenecks, enabling interactive frame rates and adaptive memory footprints without compromising visible fidelity.

A foundational strategy for taming large-scale scenes involves partitioning the environment into manageable, independently optimized blocks, followed by a consolidation phase. [9] presents a pioneering divide-and-conquer approach where the scene is split into chunks that are trained in parallel, then merged into a unified hierarchy. A key insight from this work is the adaption of the training procedure to handle the sparse and often overlapping coverage common in large captures, preventing overfitting and artifacts and ensuring visually coherent transitions between chunks. This two-stage approach—-independent training followed by hierarchical consolidation—avoids the memory explosion associated with global optimization and establishes a robust foundation for streaming and selective rendering.

Building upon this spatial decomposition, explicit LoD frameworks have been developed to dynamically manage the level of detail based on viewing conditions. [52] introduces an LOD-structured 3D Gaussian approach that stores multi-resolution anchor points. The model dynamically selects a resolution-appropriate level of detail during rasterization, ensuring performance is less contingent on the total scene complexity and providing consistent frame rates when zooming in and out. This is achieved by using a structural hierarchy to guide where new Gaussians are placed via an efficient info-gain-guided density control. This avoids unstable training and ensures that the resulting hierarchy optimally represents the spatial distribution of the signal. This approach directly tackles the variability in the number of "active" Gaussians, shifting the rendering cost from being a function of the total scene size to being a function of the current view frustum and desired detail. The challenge remains in the design of a unified and seamless mechanism for transitioning between levels to prevent popping artifacts. This is accomplished by enforcing consistent representation of the same signal across levels and using extra regularization to ensure smooth crossfades between levels.

In contrast to per-object or per-block LoD, more recent approaches like [74] and [75] address the concept of level-of-detail from a signal-processing perspective. They focus on the fundamental cause of scale-dependent aliasing, [74] showing that a mismatch between the sampling frequency of the target view and the spatial frequency of the 3D Gaussians causes the problem. This has resulted in the essential use of a 3D smoothing filter to constrain the highest frequencies and a 2D Mip filter during the splatting process. While primarily designed to handle anti-aliasing, these methods implicitly create a form of LoD by limiting the detrimental effects of scale changes. The introduction of such implicit filtering is a pivotal step towards a more unified theory of scale-consistent rendering.

Evaluate the merits and limitations of the various optimization strategies, because the choice between, for example, a cluster-based approach that prioritizes localization and a dynamic LoD approach that optimizes per-frame workloads, often comes down to the downstream needs of the application. A key factor in render quality is the quality of the input. [40] and [39] have been essential in decreasing the dependence on a pre-computed SfM point cloud. By starting from random or progressively densified point clouds, they make the pipeline more robust and less dependent on the accuracy of SfM, and also further streamlines the training by not requiring the entire scene to be reconstructed beforehand. The consolidation of such large, multi-block and hierarchical scenes, along with the adaptation to the strict latency constraints of streaming scenarios, remains an open research area. The road ahead to implement client-edge rendering will depend on integrating the LOD hierarchy with predictive pre-fetching and robust compression

techniques, as seen in the development of methods like [9] and [76], yet a truly integrated, rate-distortion-optimized solution for arbitrary scale and bandwidth is still in its infancy.

#### 4.4 Accelerated Training and Distributed Optimization

The optimization of 3D Gaussian Splatting (3DGS) traditionally requires iterative gradient descent over hundreds of thousands of iterations, with a computational cost that scales steeply with the number of primitives. This subsection examines a growing body of work dedicated to accelerating the training pipeline itself, alongside distributed strategies that enable scalability to large-scale scenes and multi-GPU environments. These efforts are pivotal for transitioning 3DGS from an offline reconstruction tool to a practical method for real-time and high-throughput applications, complementing the scalability mechanisms discussed in the previous subsection by attacking the *training-time* bottleneck that remains even after scene structure has been organized into hierarchical and LoD formats.

A substantial share of research addresses the inefficiencies embedded in the adaptive density control and gradient computation. The original adaptive density control is known to be sensitive to initialization and prone to producing redundant primitives, which inflates training time. Improved densification strategies leverage pixel-aware gradients [42] to better identify where new Gaussians are needed, while others identify and resolve the issue of gradient collision to recover fine details more efficiently [41]. Complementing these, direct frequency-domain regularization [77] can guide densification to avoid the creation of unnecessary Gaussians, leading to more balanced and compact models that train faster. To reduce overall complexity, structured pruning and sampling techniques limit the number of active Gaussians, as demonstrated by Mini-Splatting, which uses blur split and depth reinitialization to reorganize primitives for constrained resourcing [19]. These refinements directly complement the hierarchical decomposition strategies from the previous subsection: just as those methods reduce the per-frame rendering cost across spatial scales, these training-time optimizations lower the per-iteration cost of **optimizing** the same primitives, making the optimization of a multi-resolution hierarchy both feasible and efficient.

Beyond algorithmic refinements, addressing the dependency on Structure-from-Motion (SfM) initialization is a critical direction for scaling training. The brittleness of SfM has motivated the exploration of random initialization and the distillation of geometric priors from neural radiance fields [78]. Highly relevant is the development of global-local depth normalization, which can guide the optimization toward well-shaped scene geometry in a sparse-view context, ensuring accurate geometry with significantly fewer steps than standard methods [21]. Prior-based approaches, such as using a monocular depth estimator to guide the optimization, directly reduce the time required to eliminate noise and produce a clean structure [79]. These methods shift the optimization from a purely learning-based regression to a more constrained problem, drastically reducing data requirements and training steps. This shift is particularly synergistic with the offline-computed hierarchical structures of the previous subsection, since a robust initialization allows the training of block-wise partitions to converge independently and consistently, avoiding expensive global reparations during merging.

For scaling to unbounded scenes, pre-partitioning alone—as covered earlier—addresses memory scalability, but it does not inherently reduce per-block training time. Distributed training across devices is therefore paramount. The most common approach follows a divide-and-conquer block-wise decomposition [80], partitioning a large scene into manageable chunks and training each independently before fusing them; this is often coupled with novel distributed optimization techniques such as greedy local-global and ADMM to stabilize the scene and consolidate cross-block information. This spatial parallelization is a natural extension of the hierarchical scene representations from the previous subsection, which typically assume a block-wise or LoD-wise organization by construction. In contrast to data-parallelism, spatial parallelism using non-blocking and overlapping communication offers a promising direction: a non-overlapping block order minimizes synchronization bottlenecks, allowing sub-scenes to be

optimized concurrently with limited overhead. Another notable direction, exemplified by [80], takes a fundamentally different route, generating a globally aligned coherent point cloud from a pre-trained dense stereo model to bypass independent optimization and seamless integration. This approach avoids costly SfM pre-processing and demonstrates that joint quadrature can be achieved in under a minute when a dense geometric prior is used as initialization.

A further important vector for convergence acceleration is the reduction of redundant computation in the gradient and in the number of training views. Methods that distill depth information directly into the Gaussian optimization or that remove cost-historical SfM preprocessing allow for pipeline steps that would otherwise introduce overhead to be dropped. In distributed settings, reconciling data distribution shifts across nodes—if handled with careful balancing—can yield near-linear speedups. Similarly, the adaptation of the rasterizer to specific hardware, when used as a module in end-to-end scenes, can further accelerate the overall system. These steps dovetail with the streaming and client-edge scene-graph insights from the previous subsection, where low-latency training is not a luxury, but a prerequisite for immediate and interactive model updates.

In summary, the trajectory of training acceleration in 3DGS is moving from purely algorithmic and heuristic-level refinements toward more fundamental approaches that leverage geometric priors [39]. The realization that the initial point cloud heavily influences the optimization path drives the shift from SfM toward more robust and learned initializations [81]. For distributed optimization, the consensus is to build hierarchies and decompositions that allow spatial locality and parallel processing. Combined with the hierarchical scene organization and level-of-detail control from the previous subsection, these training-side optimizations set the stage for the central remaining challenge: coherently rendering these models with dynamic compression, real-time refinement, and band-limited updates—a topic that will be further elaborated in the next subsections, where we explore integration of these training-time strategies with the deployment constraints of streaming and remote rendering.

## 5 Geometric and Physically-Based Rendering

### 5.1 Decomposed Scene Representation for Inverse Rendering

The transition from photorealistic novel-view synthesis to physically-based inverse rendering necessitates a fundamental shift in how 3D Gaussian Splatting (3DGS) primitives are parameterized. The vanilla representation [7] optimizes view-dependent appearance via spherical harmonics, which entangles the intrinsic material reflectance of a scene with the extrinsic illumination under which it was captured. Decomposed scene representations address this ill-posed problem by augmenting each Gaussian primitive with additional physical attributes—namely surface normals, Bidirectional Reflectance Distribution Function (BRDF) parameters (e.g., base color, roughness, metallic), and a representation of incident light—that are optimized jointly to factorize the observed radiance.

A foundational line of work in this direction augments each primitive with both geometry and material attributes. For instance, surface normals are often estimated via the orientation of flattened ellipsoids or by gradient-based methods, providing an essential constraint that regularizes the shape and facilitates the subsequent prediction of shading parameters. By attaching spatially-varying BRDF properties to each Gaussian, methods such as those described in [82] and related approaches [83] enable the recovery of intrinsic albedo, roughness, and specularities that are decoupled from the lighting environment.

A second core approach factors the outgoing radiance directly. View-dependent effects are modeled as a function of a local illumination field and the surface reflectance, rather than by high-frequency spherical harmonics. This often involves representing the environment light with analytical models such as Spherical Gaussians (SGs) or factorized tensors, which are jointly

optimized with per-primitive BRDF features. This approach, as seen in techniques like [22] and the broader framework of [7], enhances the ability to model detailed anisotropic reflections and specular highlights.

To achieve plausible relighting under novel illumination, rendering methods must account for global light transport, including occlusions and inter-reflections. Techniques such as [60] and [23] discuss the challenge of baking visibility or handling dynamic interactions. More advanced methods, like [22] and the physically-based approach in [84], employ ray-tracing using bounding volume hierarchies or precomputed radiance transfer to estimate self-shadowing, yielding synthesis with more plausible shadows without sacrificing the real-time performance of the point-based renderer.

For scenes captured with "in-the-wild" images, the decomposition must extend to the scene-level illumination map. The methodology adopts a two-tier model: estimating a global environment map for distant light, alongside local lighting effects, as presented in [85] and [69]. This decomposition is crucial for applications like object insertion, where the novel object must exhibit lighting consistent with the surrounding environment.

In conclusion, the literature [26] [29] demonstrates a significant push from direct appearance cloning toward physically grounded representations. Yet, critical challenges remain: the ill-posed nature of the decomposition requires strong priors to disentangle materials from geometry; the computational cost of global illumination is high; and current methods often fail to capture dispersion or high-dynamic-range phenomena. Future innovation will likely focus on integrating more sophisticated neural light transport [5], and on integrating manifold regularization to handle ambiguity in the view optimization process.

## 5.2 Surface and Normal Extraction from Gaussian Representations

The unstructured, volumetric nature of Gaussian primitives presents a fundamental obstacle to extracting the explicit surfaces required for physically-based shading, mesh-based asset pipelines, and downstream geometric reasoning. Unlike implicit representations that can leverage level-set extraction, 3DGS defines occupancy through a soft blending of anisotropic ellipsoids, with no inherent notion of a continuous surface. This subsection examines the methodologies that have emerged to bridge this representational gap, focusing on two complementary paths: the estimation of normals for geometric regularization and the structural adaptation of Gaussians to become inherently surface-aligned.

The most direct approach towards geometric fidelity is the estimation of per-primitive surface normals, a task complicated by the statistical nature of the field. One line of work computes the normal as a property derived from the gradient of the local density field [37]. This requires a continuous reinterpretation of the Gaussian volume, which can be achieved by leveraging the implicit signed distance field (SDF) transformed into opacity values, thereby aligning the surface from which normals are extracted with the discrete Gaussians [37]. Methods such as GaussianShader introduce a delicate loss design that encourages alignment between the inferred normal direction and the geometric shape of the Gaussian spheres, showing that a consistent shading model can drive the learning of plausible geometry [63]. However, the inherent anisotropy of a generic Gaussian ellipsoid makes defining a single "surface normal" ambiguous. Without structural priors, the estimated normals are ill-defined and prone to noise, often failing to align with the actual physical surface geometry. This consequence often manifests as "fluffy" or semi-transparent Gaussian cloud models that misrepresent geometrical thickness, a point emphasized by representations that rely on volumetric primitives which suffer from similar limitations in capturing fine geometry [28].

To overcome the ambiguity of a generic 3D Gaussian ellipsoid, a significant line of research collapses the 3D volume to 2D primitives. Planar 2D Gaussian kernels, or surfels, provide a view-consistent geometric definition where each primitive has a clear normal that defines its orientation in space [20]. This deliberate structural change transforms the optimization challenge:

the process becomes a surface fitting problem rather than a mere density learning task. Work in this direction often combines perspective-accurate splatting with depth distortion and normal consistency terms, ensuring that the optimization results in tangency with the underlying surface and the recovery of thin structures [20]. Gaussian Opacity Fields builds on this by deriving the surface normal from the intersection ray and the primitive, which enables applying normal regularization at that specific point, significantly enhancing the extracted geometry [60]. This stable access to surface information is not just a theoretical nicety; it is a crucial operation for integrating a model with advanced shading: the normal at the surface point is required to compute the incident reflection direction, parameterize physically-based materials such as microfacet model or included specular, to implement algorithms under which the accurate normal must be made available to the physically plausible relighting pipeline, thus directly bridging the gap to the appearance modeling demonstrated in the previous subsection.

Building upon this stable access to surface information, methods quantify consistency between depth and normals to ensure physical fidelity. 2D Gaussian Splatting has established a method combining perspective-consistent 2D disc rendering with a depth distortion loss, and a normal consistency term that aligns the gradient field of rendered depth to the primitive normals, preserving detailed geometry [20]. Once normals are available, they serve as a strong supervisory signal. DNGaussian starts from estimated 3D Gaussians to truly place and constrain them in depth by applying hard and soft depth regularization terms to ensure positional coherence under monocular depth supervision, leading to a better global constraint [21]. The multi-view distortions caused by low displacement are thus remedied by a prior that emphasizes both the global shape and the local normal shape in low-variance degree. This is particularly relevant for weakly textured surface regions or open-space input, where optimizing the normals may be the sole source of guidance, also shown in [37].

Despite these advances, several challenges persist. The first is one of continuity: imposing a deterministic "direction" on the flattened or displaced Gaussian fields does not guarantee topologically correct closed surfaces, leaving "bubbles" or disconnected surfels in the 3D. In contrast to meshes, Gaussians represent an optimization over an assignment of scale; if a 3D Gaussian is spuriously near the surface, its volume state creates an ambiguous "dual-surface" configuration. This underscores the importance of robust adaptive density control. As noted in recent analysis, the naive clone-and-split based supervision of the Gaussian field does not fit the underlying physical continuity unless a careful density splatting is implemented [43, 41, 42]. Future methods must explicitly couple physical boundary conditions and oriented sampling on a coherent surface, not just for the sake of geometry, but to enable the physically based appearance decomposition discussed in the previous subsection.

The integration provided by these stable normals is enabling: it transforms an unstructured "point cloud with appearance" into a geometry that supports physically-based lighting, and it is becoming a standard component in many existing approaches, from reflective surface rendering [63] to the appearance models discussed earlier. This stable geometry foundation is what allows the whole system to accurately decompose specular reflections and intrinsic material properties. The next generation of research will likely leverage a deeper interplay between geometry and its photometric covariates, extracting stable surface priors by learning the anisotropic parameters of the BRDF and its link to global spatial and orientation context, thus creating a future where the geometry and appearance are not merely concurrently estimated but are jointly, physically coupled without losing this architecture.

### 5.3 Advanced Appearance Modeling for Specular and High-Frequency Effects

The pursuit of photorealistic rendering in 3D Gaussian Splatting (3DGS) is fundamentally challenged by the appearance model's capacity to represent high-frequency and anisotropic light interactions. The base formulation, which relies on low-order spherical harmonics (SH) to encode radiance, is intrinsically limited. SH coefficients, particularly at the low degrees typically used,



smooth angular variations and are singularly unsuited for capturing dramatic specular highlights, glints, and anisotropic reflections that are characteristic of non-Lambertian materials. As such, a significant body of research has emerged that focuses on decoupling appearance from geometry and introducing more expressive shading models that are specifically designed to accommodate the complex light-matter interactions inherent to specular surfaces.

A leading strategy to overcome the spectral limitations of SH is the substitution of the appearance basis with more adaptable functions. In particular, anisotropic spherical Gaussians (ASG) have proven to be a powerful alternative. The Spec-Gaussian method [22] replaces SH with an ASG appearance field in three-dimensional space, fancying the alignment of anisotropic properties. This approach, when combined with a coarse-to-fine training strategy, allows the model to faithfully represent specular components without a proliferation of Gaussian primitives. Beyond ASGs, the approach of deferred shading offers another path. Instead of directly attaching lighting computations to every Gaussian, the GaussianShader [63] applies shading to these primitives, while 3D Gaussian Splatting with deferred reflection system manages to model specular answer [86] by deferring the shading based on correct normal estimation.

A second class of methods attacks the problem from a different angle, introducing new structures to handle appearance complexity. For example, RadSplat [8] allows for the integration of radiance fields priors that can handle higher-order effects and scale to more complex scenes, while maintaining quality and robustness in challenging environments.

The concept of binding texture maps (or other bespoke 2D functions) to the Gaussian primitives is a developing field, and some works could the explicit geometry as the proxy to manage spatially varying high-frequency attributes, which are often only weakly modeled by global basis functions.

**Whereas the above-mentioned methods reduce the model’s demand for a lot of parameters, the state-of-the-art is now moving towards hybrid neural-appearance. These new methods bake the appearance former during the optimization, but this often leads to extra cost. The difficulty, as is normally the case, remains in the absence of a strict and good physical grounding for the whole optimization, which can cause appearance property variables to be combined in a way that is not physically plausible.**

Some works have found a stable way to express lighting with exposure models, but often at the cost of ignoring the dynamic reflections, which may not allow to model physical reflectance correctly in every case.

In future, work that unifies appearance with a shape consistent with the corresponding **bidirectional scattering distribution function (BSDF)** set of laws, such as [87] which has shown achievements in geometry and material recovery, likely to push the field forward. They will make the full fidelity to the scene more explainable, breaking the field away from the inherent limitation of SH.

A survey of future directions may point to solving the remaining conflicts in a unified framework that aims to amend the lighting with high-dimensional semantic representation for the physics of materials and an explicit geometric foundation, enabling texture and specular correctness in real-time.

## 5.4 Relighting and Global Illumination in Gaussian Splatting

The advent of physically-based relighting within the 3D Gaussian Splatting (3DGS) framework marks a significant departure from the original model’s reliance on view-dependent spherical harmonics, which conflate material reflectance with incident illumination. The primary challenge in this domain is to decompose the observed radiance of a scene into its intrinsic components—geometry, material properties, and lighting—to enable the re-rendering of the scene under novel illumination. This decomposition is fundamentally ill-posed, and the explicit, point-based nature of 3DGS introduces unique degrees of freedom that can lead to geometry-appearance

ambiguity, a theme that has been a recurring obstacle throughout the previous subsections. Early approaches sought to regularize this by integrating strong geometric priors, for example, using differentiable sphere tracing with signed distance functions to provide a more stable geometric foundation for material estimation [34, 47]. The move to explicit Gaussian primitives, however, offers the advantage of a direct, structured parameterization that can be more easily coupled with physically-based rendering (PBR) models, a direct benefit of the surface-aware geometric refinements previously discussed.

A prevailing strategy for tackling the inverse rendering problem within the 3DGS domain is to disentangle the appearance of each Gaussian into physically meaningful attributes. Early influential work in implicit domains established the precedent of using parametric models for efficient light transport, such as representing specular BRDFs and environmental illumination with mixtures of Spherical Gaussians (SGs) [88]. This direction has been extensively explored to render specular reflections and materials, albeit often at the cost of architectural modifications that increase the computational load. The shift to the 3DGS domain has seen the adoption of similar strategies, targeting real-time and high-quality inverse rendering. Techniques in this vein involve jointly optimizing per-Gaussian normals, material parameters (such as base color, roughness, and metallic properties), and a representation of the incident lighting. A key challenge here is estimating accurate surface normals from the discrete and volumetric distribution of semi-transparent Gaussians, as the specular term is highly sensitive to normal orientation, which connects directly to the lightweight appearance models discussed in the previous subsection. Line-of-sight continuity remains a central research theme, where the local planning of the material properties against the splatting process can lead to more coherent geometry and highlight behavior in the presence of specular and non-Lambertian surfaces [88, 89].

While pure decomposition stores the illumination in a fixed global model, achieving true global illumination (GI) requires a mechanism for dynamic light transport. To simulate the complex interaction of light—including shadows, specular reflections, and interreflections—the rendering pipeline must account for light paths that are not directly visible from the camera. There are two main computational pathways to address this challenge within 3DGS. One pathway is to integrate with hardware-accelerated ray-tracing APIs and build an explicit acceleration structure over the Gaussians, which allows for sampling visibility from a light source and on-the-fly computation of shadows and reflections, moving beyond the earlier uniform ray-casting approaches [90] and allowing for more versatile gradient acquisition. Another compelling pathway is the baking of global illumination effects into the Gaussian primitives themselves, effectively resulting in a precomputed radiance transfer (PRT) framework. In such an approach, parameters like visibility and indirect light bounce coefficients can be stored alongside colors; these latent coefficients are learned during training and can then be coupled with a blended lighting model for real-time relighting. The benefit of a PRT-style model is clear for scenes with static geometry or interactive lighting updates, where it provides a direct and cost-efficient solution with continuous path tracing in a real-time setting, and it offers a practical sense of physical realism, especially under dynamic light.

A notable practical challenge for relighting is the high dynamic range (HDR) of scenes, as light goes far beyond the 8-bit range of typical imagery. Standard three-channel LDR images can only capture a limited amount of information, often over-exposing the highlights and under-exposing the shadows. The physical correctness of relighting arguably relies on operating in the HDR domain; however, the standard 3DGS is trained on, and is deeply entangled with, LDR inputs. This limitation opens a promising future direction for addressing the inverse problem in a generative manner, where recent advances in diffusion-based models have been leveraged to hallucinate new views and lights for scenes in 3D, compensating for the lack of accurate geometry and material cues [91, 92]. However, the accuracy for precise "correct" relighting of materials requires strict physical understanding of the illumination, which leads to an inherent trade-off: the fidelity of physically-designed models in inverted scenes versus the flexibility

of generative hallucination. The field is therefore moving towards the construction of hybrid approaches: methods that wisely invert the scene’s intrinsic complexity or fundamentally rest on an iterative abstraction for physically-based mechanisms, enabling a spectrum of solutions that are adaptable, globally lit, and aesthetically pleasing, bridging the gap between captured appearance and actionable space for both artists and engineers.

## 5.5 Challenges and Opportunities in Material-Specific Appearance

The pursuit of physically-based rendering within the 3D Gaussian Splatting (3DGS) framework confronts a fundamental paradox: the representation’s strength—its discrete, view-dependent splatting of radiance—is simultaneously its primary obstacle to modeling complex, material-specific appearance. While the explicit nature of Gaussians facilitates fast rendering, it lacks the inherent surface parameterization and the global light transport awareness needed to disentangle a scene’s intrinsic material properties (albedo, roughness, metallicity) from its extrinsic illumination. This disentanglement is the central challenge, as the optimization of geometry and material attributes from sparse multi-view inputs is severely under-constrained. Methods that attempt to jointly estimate normals, BRDF parameters, and lighting, as explored in frameworks like Relightable 3D Gaussian and inverse rendering pipelines [82, 7], often suffer from a coupled ambiguity, where errors in geometry are compensated by inaccurate material estimates and vice versa. A key opportunity lies in the use of monocular priors and multi-view photometric consistency, as seen in PGSR and DN-Splatter, to enforce disentanglement and guide the optimization toward geometrically plausible solutions [36, 35].

A second critical obstacle is the limited expressiveness of the standard appearance model in the original 3DGS. The reliance on low-order spherical harmonics (SH) to represent view-dependent color is well-documented for failing to capture high-frequency specular reflections and anisotropic highlights. This is particularly problematic for materials like metals, glass, and glossy plastics, where subtle view-dependent changes are integral to their visual identity. To address this, Spec-Gaussian introduces anisotropic spherical Gaussians (ASG) to encode the appearance field of each 3D Gaussian, significantly improving the ability to model specular and anisotropic components without increasing the primitive count [22]. Similarly, Mixture of Volumetric Primitives (MVP) and other hybrid approaches hint at the potential for more flexible appearance models [28]. This quest for higher-fidelity appearance modeling creates a critical trade-off between computational efficiency and visual accuracy, as more expressive methods like ASG or neural network-based decoders increase both training time and memory footprint, directly conflicting with the core real-time rendering promise of 3DGS.

The efficient simulation of light transport—specifically, providing plausible shadows, inter-reflections, and transparency—remains a significant hurdle. Directly computing global illumination for millions of Gaussian primitives at interactive rates is computationally infeasible. While some methods have proposed precomputed radiance transfer or visibility baking to capture static global illumination effects, these approaches struggle with dynamic scenes or relighting scenarios. Furthermore, despite the potential of a Gaussian representation for point-based rendering, as shown in TRIPS: Trilinear Point Splatting for Real-Time Radiance Field Rendering, refractive or translucent effects require non-trivial light paths not readily supported by the rasterization pipeline. This narrows the practical applicability of 3DGS in, for example, product visualization for medical applications. Finally, the challenge of capturing and representing a wide dynamic range of illumination is notable, as the standard optimization pipeline is not designed to handle physically accurate tone mapping or linear light transfer—a crucial limitation for scenes with concentrated light sources or dynamic lighting environments.

Looking ahead, several promising future avenues emerge. The development of more robust and physically grounded regularization terms is crucial for robustly disentangling geometry, material, and lighting. Additionally, advancements in efficient GPU-based ray tracing could enable real-time global illumination for point-based primitives, unlocking a new level of realism for

relighting. Finally, exploring hybrid representations that integrate the explicit 3DGS primitives with implicit fields for light transport could offer a path forward, as seen in early explorations like GSDF [69]. Ultimately, addressing these material-specific challenges will be a definitive test of 3DGS’s ability to mature from a novel view synthesis tool into a true 3D representation for a wide range of graphics and learning applications, including generative models that require the proper coupling of geometry and appearance [93, 26].

## 6 Generative Modeling and Content Creation

### 6.1 Score Distillation-based Generation from Text and Images

Score distillation has emerged as the dominant paradigm for zero-shot text-to-3D and image-to-3D generation, leveraging the rich priors embedded in large-scale pre-trained 2D diffusion models to guide the optimization of 3D representations. The foundational formulation, Score Distillation Sampling (SDS), was initially developed for implicit representations such as NeRF and operates by perturbing a rendered image with random noise and computing a gradient signal that pushes the rendered distribution toward the learned prior distribution of the diffusion model. The advent of 3D Gaussian Splatting (3DGS) introduced a fundamentally different optimization landscape: explicit, unstructured primitives whose parameters can be directly manipulated, significantly accelerating optimization and permitting real-time rendering during the distillation process. Consequently, methods like GSGEN [16] adapted SDS to the Gaussian representation, exploiting its explicit nature to employ a progressive optimization strategy. This strategy first establishes a coarse, 3D-consistent geometry using point cloud diffusion priors alongside SDS, and then applies a compactness-based densification process to refine appearance [16]—an order of operations that offers crucial benefits. The explicit point structure of early-stage Gaussians, underpinned by point cloud diffusion priors [94], is critical for mitigating the Janus problem (multi-face artifact) that plagues implicit representations for text-to-3D generation [16], while the subsequent dense refinement stage enables the capture of high-frequency detail. This hybrid approach underscores the central thesis that by moving beyond purely implicit coordinates to explicit primitives, the optimizer can integrate independent geometric 3D priors, not just 2D rendered views.

However, the initial naive application of SDS to 3DGS immediately collides with a well-documented failure mode: multi-view inconsistency and over-smoothing. The primary issue stems from the inherent uniform noise strategy within SDS, which pushes the crafted scene towards a model-averaged, over-smoothed state, thereby providing ambiguous or unrealistic capabilities. To address these constraints, a spectrum of distillation variants and regularizers have been integrated. Variational Score Distillation (VSD) frames the pursuit as a variational inference problem, where the optimizer seeks a distribution over 3D assets controlling the discrepancy from the diffusion posterior. This variant has proven essential when synthesizing high-frequency texture on an initial geometric estimate. Alongside VSD, asynchronous and annealed timestep scheduling has become key: starting with higher noise levels matches the rendering to a coarse low-frequency structure before falling into finer details, avoiding premature Gaussian splitting and densification in early phases. These prior-guided and time-scaled components mark the shift from naive SDS to robust, multi-stage pipelines for generation.

Addressing the specific geometric inaccuracies necessitates a two-fold remedy. First, per-camera prompt guidance has emerged as a simple yet effective mechanism. By concatenating both the global text prompt and view-specific part prompts—such as "front view" or "back view"—the distortion of 2D prior to misinterpreting multiple viewpoints from a uniform textual description is alleviated. A more intricate, and therefore powerful, test is the integration of attention refinement into the SDS/diffusion critic. As the diffusion UNet’s cross-attention layers hold rich, geometry-aware semantic information (e.g., a mapping between object parts and

text words), the edited generator can use these as a geometric prior. By using cross-frame attention maps in the Gaussian rendering branch, coupled with a prompt-aligned attention loss, the network also receives a well-structured 3D layout from the 2D generator about where each component must be located, preventing texture bleed and shapes under long-horizon prompts. Using pipelines like GSGen [16], these blocks are successfully shown to halve the presence of the Janus artifact when measured across multiple frames, proving the necessity of structured 3D information injection.

Beyond mere structural guidance, the need for richer regularization surfaces in the objective function as orthogonal constraints. The problem is fundamentally under-constrained: thousands of Gaussians are being optimized from a density of signals that only exists as re-rendered gradients from a black-box diffusion model. A blended objective function is therefore common practice in parameterized synthesis. A seminal line is to introduce the score distillation module as a diffusion process itself, in which either the rendered image or its gradients are themselves fed into a second denoising process to output content from valid, coherent target distributions. Such formulations align the predicted canonical view with text. This inherent balance reinforces the view that the value of the Score Distillation technique will come from methodical composition, not any single elegant formula.

Future advancements in score distillation for Gaussian scenes should be searched for in unstructured, multi-scenes and direct-distillation of features into the explicit primitive layout. Rather than optimizing via billions of differentiable camera rays that backpropagate through alpha-blending, the splat representation grafts well onto learned, latent-space dictionaries that encode practical geometry and contextual priors [93]. The dual movement from image-prior critics to 3D-prior as well the Gaussian parameter-spaces suggest enormous room for semantically grounded edits and generalized control from sparse views—paving the way for Gaussian content creation from natural language that intrinsically matches the core appeal of the field: perception and reality at an interactive frame rate.

## 6.2 Feed-Forward and Large Reconstruction Models

The paradigm of per-scene optimization, while delivering exceptional fidelity, imposes a fundamental computational bottleneck that precludes real-time interactive applications and large-scale content generation. Feed-forward and large reconstruction models (LRMs) address this limitation by learning a direct mapping from observations to 3D Gaussian parameters, thereby enabling the synthesis of 3D assets in a single forward pass without iterative refinement. This shift from optimization-based to regression-based reconstruction constitutes a critical step toward scalable and democratized 3D content creation, complementing the generative and score-distillation approaches discussed earlier by offering a path to instant, amortized inference that does not require per-instance optimization loops.

Central to this approach is the estimation of a structured set of Gaussians from sparse or even a single view, a problem that leverages the explicit primitive representation to directly predict scene geometry and appearance. A pioneering strategy employs cost-volume-based architectures that compute a geometric cost volume from multi-view images to infer per-pixel depths, which are subsequently unprojected into a 3D Gaussian point cloud. This geometric grounding ensures spatial accuracy and multi-view consistency, avoiding the ambiguity inherent in purely appearance-based regression. Concurrently, triplane-Gaussian hybrid architectures have emerged as a powerful alternative for the more challenging single-view setting. By decoding a triplane feature representation with a hybrid 2D-3D network, these models efficiently regress a dense set of Gaussian attributes, directly leveraging spatial locality to generate faithful 3D assets in under a second—a speed advantage that, when coupled with the rendering efficiency of splatting, makes them particularly suited for interactive applications.

To ensure global structural coherence, especially for objects with complex topology, further architectural innovations incorporate structural supervision and global attention mechanisms

[95]. A large reconstruction model based on a hypergraph neural network (HGNN) and an attention-based transformer backbone facilitates 3D-aware reconstruction by reasoning over patch-level relationships, preserving the geometry of complex objects across all camera views [95]. However, a critical bottleneck in many of these feed-forward models is the reliance on a regression loss defined on intermediate representations, such as dense depth maps. This poses a quantitative challenge: an accurate depth map does not directly guarantee a photorealistic or texture-complete reconstruction, as appearance details are often lost in the process. To directly optimize for the final visual output, a novel architecture regresses both 3D Gaussians and surface normals, with the joint prediction integrated into a differentiable splatting-based rasterizer. By applying a photometric loss directly to the rendered image, this method optimizes appearance in an end-to-end fashion, bypassing the need for intermediate supervision and improving accuracy on unseen data—an end-to-end objective that parallels the score-distillation losses used in generation, yet avoids the heavy per-scene computation.

The frontier of this field is being pushed by zero-shot large multi-view reconstruction models, which leverage cross-view attention to extract global features from an arbitrary number of input images. These generalizable decoders establish an effective correspondence mechanism to directly regress Gaussian parameters, supporting fine-grained custom generative tasks and significantly advancing the frontier of generalizable and scalable 3D reconstruction. The choice of structure for the output Gaussians is also a central design consideration; while most models predict a fixed number of independent primitives, more sophisticated approaches predict a structured form—for example, arranging Gaussians into a predefined dense voxel grid via optimal transport [95]—this enables the adoption of standardized 3D backbone networks (such as 3D U-Nets) in diffusion models, bridging the gap between generative algorithms and the underlying splat representation, coherently linking back to the generative paradigms above.

Despite their impressive speed, the quality of feed-forward models—such as those relying on a video diffusion model to generate multi-view consistent images and reconstruct the 3D shape in feed-forward [95], and those that predict the entire underlying set of 3D Gaussians—runnettedly in a feed-forward manner—remains constrained by the representational capacity of the Gaussian parameters and the fidelity of the final output; they often generate coarser geometric details per scene-optimization methods. Future investigations into joint appearance and geometry learning, image-based compositing, and the integration of more powerful 2D foundation models (as explored in the following subsection) will be essential to seal this gap and unlock the full potential of feed-forward generation.

**Refinement Notes** (not part of the final answer): The refined version preserves all original citations, maintains the core technical content, and improves coherence by: (1) introducing the subsection with a clear thematic link that references the optimization-based paradigm discussed in preceding subsections and sets up the feed-forward transition, (2) smoothing transitions between paragraphs (e.g., explicitly linking structure design to generative models and to the following subsection), (3) clarifying the contrasts and parallels with previous sections, (4) eliminating redundancy while preserving all technical details, and (5) ensuring that the final paragraph directs the reader to the subsequent subsection on generative output. The citation markers remain unchanged and in place.

### 6.3 Generative Models with 3D Gaussians as Intermediate Representation

The integration of 3D Gaussian primitives as an explicit, rendering-ready output within generative architectures marks a paradigm shift from optimization-based reconstruction to direct, feed-forward content synthesis. Unlike implicit representations that require volumetric ray marching, Gaussians offer a differentiable, rasterization-friendly interface that is naturally compatible with various generative frameworks. This subsection explores how GANs, diffusion models, and autoencoding architectures have been re-engineered to output Gaussian parameters, enabling real-time, editable, and high-fidelity 3D generation from diverse conditioning signals.

Generative Adversarial Networks (GANs) provide a blueprint for several Gaussian-centric generation pipelines. AGG [96] presents an amortized generative framework that directly produces 3D Gaussians from a single image, utilizing an intermediate hybrid representation with both geometric and appearance attributes. By decomposing the generation of Gaussian positions and auxiliary attributes into a joint optimization loop, AGG forgoes expensive per-instance optimization steps inherent in score-distillation methods. A cascaded super-resolution module further enhances the coarse output of the generator, producing high-quality guarantees for the explicit structure of the final 3D model. This attention to stable Gaussian placement and appearance estimation reduces learning variance and allows for highly efficient, one-shot assets, which quantifiably outperform optimization-based Gaussian frameworks in speed and competitive generation quality.

Within latent diffusion paradigms, a novel class of autoencoders employs Gaussian representations in 3D latent spaces to mediate between deterministic encodings and probabilistic generative distributions. For instance, latentSplat [97] advances a line of variational Gaussian modeling by incorporating uncertainty directly into the latent variables. Instead of predicting deterministic points, this method encodes visual scenes as a set of variational 3D feature Gaussians, allowing the representation to efficiently capture the inherent ambiguity in 3D reconstruction. Trained on video data, latentSplat demonstrates how a variational approach can be used to sample Gaussian parameters, enabling generalizable 3D reconstruction that can synthesize novel high-resolution views in real-time, avoiding the strict reliance on heavy volumetric rendering or per-scene optimization. This approach bridges reconstruction and generation by making the base Gaussian features an explicit stochastic, splattable medium, which enables not only novel view synthesis but also generative editing in a probabilistic space. Furthermore, feed-forward generalizable reconstruction models such as pixelSplat [98] have demonstrated a substantial effectiveness of feed-forward approaches, predicting multi-modal probability distributions over point depth from image pairs. pixelSplat bypass thereafter test-time optimization, showing that dense 3D reconstruction can be performed via direct feed-forward regression disposition on high-quality geometric accuracy.

Across these diverse paradigms, an important synergy emerges. While GAN-based methods prioritize fidelity and training stability with adversarial objective terms, diffusion and variational approaches champion the ability to characterize the ambiguity and uncertainty of the generated scene [97]. Hybrid strategies that fuse the speed of amortized Gaussian prediction with the probabilistic expressive power of diffusion [97] [96] represent a promising future frontier [29, 26]. Further progress will likely hinge on inventing more expressive and physically-informed Gaussian parameterizations and the creative implementation of adversarial and multi-level diffusion schemes that can be efficiently decoded by the rasterizer.

## 6.4 Scene Editing, Stylization, and Manipulation with Generative Priors

The arrival of generative priors—particularly large-scale pre-trained diffusion models—has fundamentally shifted the paradigm for editing and manipulating reconstructed 3D scenes. For 3D Gaussian Splatting (3DGS), the explicit, point-based nature of the representation provides a unique advantage: editing can be performed directly on the primitives, and the consequences can be visualized in real time. This synergy has created a rich body of work on text-driven editing, style transfer, and generative inpainting, all anchored by the central challenge of ensuring multi-view consistency in the edited output. While the generative methods discussed in the previous subsection synthesize new content from scratch, the editing approaches presented here operate on already-reconstructed scenes, leveraging the same explicit Gaussian parameterization to enable post-hoc manipulation with unprecedented speed and control.

The most direct approach to text-driven editing leverages the InstructPix2Pix framework to iteratively modify the training images of a scene, then re-optimizes the 3D representation to match the edited set. While this was successfully demonstrated for NeRFs [99], its application

to 3DGS is more challenging due to the inherently high-frequency, pixel-level detail of these 2D image edits clashing with the global 3D optimization process. To address this, methods like [17] exploit the explicit nature of 3DGS by introducing a Gaussian semantic tracing technique that identifies Gaussian primitives corresponding to the target editing region. This spatial localization, combined with a Hierarchical Gaussian splatting mechanism to stabilize the optimization under the stochastic guidance of diffusion models, allows for precise, localized edits of color and geometry without affecting the rest of the scene. Similarly, [67] proposes an iterative paradigm that alternates between a 2D diffusion model and the 3DGS optimization, allowing for more delicate and controlled modifications. However, a limitation of these iterative methods is that they can be slow, as each edit requires a full re-optimization.

A more recent and efficient departure from the image-by-image editing approach is the use of depth-conditioned and cross-view attention. The [100] method tackles the computational bottleneck by editing a collection of rendered images simultaneously as a batch, rather than iteratively. The core innovation is to enforce geometric and appearance consistency through a depth-conditioned diffusion model, which leverages the naturally consistent geometric depth maps from the 3DGS scene to constrain the edits in 3D space. To further unify the appearance of all edited views, it employs a cross-view attention mechanism that aligns latent codes across multiple reference views, effectively sharing the stylistic and textural changes globally. This results in significantly faster editing speed and higher visual quality, especially for scenes where a global change is requested. For more aggressive, original style rendering, [101], originally designed for NeRF, demonstrates a more robust method using a nearest-neighbor-based loss to capture the artistic texture and brushstrokes from a reference style image. This loss functions at the level of a full-resolution image rather than a patch level, avoiding overly blurry results that Gram-matrix-based losses often produce. Although this is a general method, its formulation as a radiance field requires careful adaptation for 3DGS—a transfer that leverages the explicit geometry of splats to achieve similar artistic control without the computational overhead of ray marching.

Beyond style, common operations for removing unwanted objects also require generative models. The [102] architecture constructs a 3D inpainting pipeline by using sparse annotations from a single input image, followed by an optimization-based approach that leverages learned 2D inpainting, combining the two objectives of preserving the scene content and achieving multi-view coherence. When applied to 3DGS, the removal of the target object requires the output from the 2D inpainters to be distilled into the 3D space. This process maximizes accuracy and can be done efficiently by exploiting the explicit geometry and the rasterization engine of Gaussian Splats, making the iterative refinement of the edited region converge far faster than in implicit representations.

For semantic editing that requires understanding of object identity and category, the integration of large-scale vision-language foundation models into 3DGS has proven essential. Methods like [103] infuse the scene with dense semantic features extracted from models like SAM and CLIP. This creates a versatile and rich editing layer, which parallels the semantic anchoring used in the generative avatar methods discussed in the next subsection. Once the 3D scene is decomposed into its semantic feature fields, the user can perform open-language object selection, segmentation, or local editing by simply positioning a 3D point or bounding box in the scene. This marks a bridge between purely visual reconstruction and high-level language, with core edits reflected in 2D sampled views through the fast final rasterization step—making interactive, language-driven manipulation of 3D content a practical reality.

A future direction for 3DGS is editing with precise geometry; most prior methods focus on appearance rather than the underlying shape of the scene. The structure of Gaussian primitives—locations, scales, rotations—can bias the geometry in a few scenarios. While a generative prior can modify the large appearance, it can also lead to corrupted geometry if not properly incorporated. To resolve this, one can harness the point-based nature of the representation



through geometric editing properties [34] to achieve accurate shape manipulation. For instance, a more robust approach integrates diffusion priors that guide both geometry and texture, as in [104], which uses a view-dependent diffusion model to sculpt geometry and texture on the fly, and subsequently employs bootstrapping stages to boost texture fidelity. This bootstrapping leads to more coherent geometry, going beyond the stylistic appearance flicker. The strength of the explicit representation is that these geometric techniques can be directly applied to the 3DGS model while enhancing generality; however, this comes at the cost of high first-processing time and unstable memory management issues.

Looking toward the future, while [67] and [67] make 3DGS editing of static scenes more accessible, dynamic scene editing remains largely open. The near-real-time, interpretation-friendly approaches discussed here—which complement the generative profile of the previous subsection—lay an effective groundwork for defining 3D-aware edits for time-varying scenes. The superiority of 3DGS also requires that style transfer and inpainting operations maintain temporal stability, processing video sequences sequentially to avoid excessive flicker. A key open problem is the adaptation of generative priors to the 4D space, ensuring that scene parameters and pixel values are consistently tied to the explicit moving Gaussians. Essentially, editing events such as “remove the object” without leaving a trace of the 3D geometry require hybrid systems where neural fields and generative diffusion models are jointly marshalled to close the gap between the statistical and geometric domains. The final frontier is to fabricate rare but deep geometry—the entire 3D topology—by linking effective control into the generation process and thereby extending localized editing to global and whole-scene modifications, a step complementary to the deformable and parametric models introduced in the following subsection.

## 6.5 Controllable Asset Generation for Virtual Humans and Avatars

The generation of controllable virtual humans and avatars represents a critical frontier in 3D Gaussian Splatting (3DGS), where the explicit, differentiable nature of the representation is uniquely positioned to address the long-standing challenges of identity fidelity, expression control, and real-time re-animation. Unlike holistic scene generation, this domain necessitates the integration of strong parametric priors to disentangle shape, appearance, and motion, effectively converting the unstructured Gaussian cloud into a semantically editable digital human. The foundational challenge lies in anchoring the typically unordered set of Gaussians to a deformable human model, integrating the expressiveness of 3DGS with the parametric structure of human models. A dominant trend involves embedding Gaussians within a canonical space governed by parametric meshes, such as SMPL-X or 3DMMs. By binding Gaussians to mesh primitives, methods such as Gaussian Mesh Splatting (GaMeS), which parameterizes each Gaussian on the vertices of a mesh face, allow for animating, rigging and sculpting by treating the underlying mesh as an efficient control mechanism [61]. Works like GaGS integrate 3D geometry features explicitly into the learning of deformation, ensuring that the learned deformation field is geometrically coherent, preventing inherent volumetric artifacts when reposing [105]. Moreover, Holistic Urban 3D Scene Understanding (HUGS) employs a hybrid "static and dynamic" 3D Gaussian representation for urban scenes, binding dynamic Gaussians to SMPL-X model parameters, demonstrating robustness to noisy 3D bounding boxes when tracking moving humans in real-time. These works emphasize the explicit representation to handle broader and dynamically reposed entities rather than just replicating.

A second major approach targets avatar generation through parametric latent spaces, focusing on maximal identity editing and explicit control. Generative models specifically learning and adversarial training provide strong identity conditioning, preserving both identity and appearance through tri-plane-based rendering. Specifically, diffusion-based tri-plane and MoE Gaussian methods navigate the trade-off between controllability and computational efficiency. The "casting" of explicit controllable Gaussian formulations, such as solidified statistics, often leads to overfitting near the 2D or 3D end of the pipeline. The primary resolution is a decoupling between

the latent basis and the plausible constraints. We see boundaries drawn from (i) tri-plane anchored latent representations, and (ii) synthesis that decouples appearance from identity, largely built around neural volumes or feed-forward and 3D-structure neural rendering; resulting in a "mixture" of full-body generation techniques.

Overall, the intrinsic smoothness and invertibility of the representation (for the whole pipeline) greatly limit both semantic and pose transformations from staying registration-clear. For fully control to become practical, the generative stage must retain both photorealism and rapid deformability across many ranges of poses. There is one striking built-in from integrating body part semantics during the relative Gaussian generation: such that semantic filters facilitate API-of-class-tight methods. The output point where this enabled a giant success of large "digital humans" is solid.

Furthermore, generated assets are often benchmarked for post-optimization. Given full user-specified rerposing or suppression of deformations, was capable as require. High-frequency editing and confounded features within the tuning. The generated Gaussians are trained slowly with the underlying mesh in the applied setting, yielding a feasible decoupling. Remaining operators can be removed.

In summary, bridging 3DGS with 3D human parametric models achieves the uniquely displayed promise: it constructs geometrically parameterized attributes, in a photo-realistic, physical appearance pipeline. While fully-generative capabilities are in their growth stage, progress is accelerating due to advances in controlling canonical topological distribution and the systematic trainability of three novel of new dimensions. The primary remaining challenge is to achieve a generalizable, truly unified digital human model; yet current innovations in the generative avatar and reenactment scheme have laid a strong stability that reshapes the entire visual effects and interactive applications.

## 6.6 Scene-Level Compositional and 360-Degree Content Creation

Scene-level generation represents the final frontier for 3D Gaussian Splatting (3DGS)-based content creation, moving beyond isolated objects and individual avatars toward holistic, navigable virtual environments. While the previous subsection focused on embedding Gaussians within parametric human models to achieve controllability over a single deformable subject, this subsection shifts the emphasis to the macroscopic inverse: how to scale generative capabilities to complete 360-degree scenes and how to compose multiple independent objects into globally consistent 3D layouts. The two dominant paradigms—panoramic generation and compositional synthesis—each address fundamental challenges inherent to scaling generative models to environmental scale, including resource constraints, viewpoint coverage, and semantic coherence across disparate spatial regions.

For 360-degree content, a prominent strategy involves the native panorama-to-3D pathway. This approach circumvents the need for multi-view diffusion models by first generating a full spherical image, which then serves as the sole, information-dense supervision signal. The integration of monocular depth estimators allows these 2D panoramas to be lifted into 3D, providing a coarse geometric structure that is subsequently refined and enhanced with a 3D Gaussian representation to enable exploratory navigation. This workflow significantly reduces the computational burden and viewpoint inconsistency issues inherently associated with directly optimizing multiple perspective views, mirroring the modular, single-subject-to-full-environment progression established in the prior subsection. However, the inherent quality ceiling of the initial 2D generation remains a bottleneck: post-hoc refinement cannot fully recover fine geometric details that are lost during the initial flat projection, just as the parametric manifolds in avatar generation impose limits on pose extrapolation. Through this pipeline, the high-dimensional distribution captured by the diffusion prior allows for the synthesis of fully immersive and unexplored, navigable 3D environments from a simple text prompt, where the final rendering quality is ultimately contingent on the fidelity of that initial panoramic image [106, 103].

For compositional synthesis, the critical problem is the multi-object, multi-environment integration: how to place independently generated objects into a cohesive 3D scene with correct geometry, scale, and semantic relationships. A common approach is the use of a single multi-view diffusion model as a critical prior, from which both the target object and the surrounding background environment are generated simultaneously. This ensures that the foreground and the scene-boundary do not **conflict** and are inherently photo-consistent, as they are guided by the same denoising trajectory [107]. This stands in contrast to stronger generative methods, where more controllable injection strategies are adopted. Such approaches employ a feed-forward model to reason about the spatial anchoring of a foreground object within a pre-existing scene, grounding the object’s position by analyzing geometric cues, then generating the object as an inpainting or insertion task. This is particularly effective for "grounded" insertion tasks—for example, placing a chair in front of a desk—where the target must be placed in a semantically and physically plausible location. Similar to how parametric mesh anchors enable semantic control over humans, this modular insertion provides user-level control over each component and is more memory-efficient than end-to-end generation, striking a balance between synthesis and spatial grounding.

Future research must move beyond the current state of "single-subject" or "object-and-scene" composition toward unifying these paradigms for true "scene-in-the-wild" generation. The primary challenge is shifting from *scene-as-assembly* creation to representing the *entire environment*—including global lighting, reflections, and inter-object interactions—as a single, cohesive 3DGS representation. Current compositional methods often treat objects in isolation, failing to capture the global illumination effects that connect them [108]. Looking ahead, the integration of physically-based material decomposition—as explored in recent relighting works—will be essential. This will enable generative models to synthesize not just geometry but also the physical properties of a scene, allowing for dynamic relighting and object insertion that automatically harmonizes with the environment’s light field. Furthermore, developing hierarchical 3D-aware generative frameworks, such as those based on Level-of-Detail (LOD) structures, will be key to achieving scalability for unbounded and traversable scenes composed of millions of Gaussians [26, 9]. This represents the natural extension of 3DGS—from a re-animated avatar to a fully interactive and immersive digital world—that is both the ambition and the final obstacle of generative 3D content creation.

## 7 Applications and Scene Understanding

### 7.1 Simultaneous Localization and Mapping (SLAM) and Dense Scene Reconstruction

The integration of 3D Gaussian Splatting (3DGS) into Simultaneous Localization and Mapping (SLAM) systems marks a paradigm shift in dense visual perception, transitioning from purely geometric reconstructions to photorealistic, differentiable scene representations that are jointly optimized with camera trajectories. This subsection examines how the explicit, point-based structure of 3DGS is uniquely suited for real-time SLAM, enabling dense geometry estimation, high-fidelity novel-view synthesis, and robust tracking within a unified framework. The survey by [18] traces the evolutionary trajectory from classical hand-crafted methods through deep learning-based approaches to the current radiance-field era, observing that discrete point representations have achieved a role analogous to landmarks used in classical visual odometry while producing continuous, differentiable maps.

The central stream of research addresses dense tracking and mapping for RGB-D sensors. Differentiable rendering of Gaussian maps facilitates the joint estimation of camera pose and map parameters by minimizing photometric and geometric residuals simultaneously. To this end, joint optical flow and depth are estimated using a direct approach that optimizes the Gaussian map and poses. In a closely related approach, a compact and fast trainable 3D Gaussian

map is used to jointly optimize camera pose and map parameters, with several designed to prune and clone Gaussians to maintain a low number of primitives while ensuring colour and depth consistency. These methods also exploit the differentiable rasterizer that differs from the original implementation by optimizing the covariance of the Gaussians to enable accurate dense geometry reconstruction and structure-aware map optimization. This progress is significant as, unlike earlier NeRF-based SLAM, these systems allow real-time tracking that matches or exceeds traditional visual odometry.

A major open challenge is long-term drift in large trajectories and loop detection. To address this, loop closure and submap registration are introduced, with a method that aligns 3D Gaussian submaps when a loop is detected using pose graph optimization to maintain global consistency. This effectively re-establishes the correct geometry and avoids the accumulation of errors over time. This strategy is advantageous compared to NeRF-based SLAM, where global loop closure is challenging due to a lack of registerable geometry or effective optimization without local minima.

Extensions beyond dense RGB-D have also been made into monocular systems. In such systems, the process relies on input poses estimated from a virtual odometry, and depth maps are estimated from monocular cues to anchor sparse inputs, making the existing robust system applicable in scenarios without a depth sensor. Upcoming work includes multimodal systems that combine semantic understanding with Gaussian representations [109]. Such hybrid systems may provide reliable localization, mapping, and semantic understanding, forming the basis for a robust next generation of robots.

These methods use the 3D Gaussian Splatting as a differentiable ray-casting representation without any surface regularizers; the map’s implicit smoothing of the uncertainty in pose and geometry is generated directly from the covariance of the Gaussian. For example, state-of-the-art methods show that a Scene Model that jointly infers uncertainties of both pose and the scene is able to distill depth noise into the covariances of the map points, leading to better tracking robustness. In summary, applying physics-based priors to the Gaussian map, such as local rigidity, or building a system with temporal configuration is fully capable of controlling scale beyond rendering practice. Yet, there remain limitations of Gaussian SLAM in dynamic and large-scale environments: using the Gaussian representation maintains mostly static scenes while being computationally limited; standard SGD optimization of the Gaussian is not designed for the large 3D base maps; the failure to prioritize memory before temporal robustness and incremental adaptation to the dynamic world, and the unprincipled mapping of uncertainty. Future 3DGS-SLAM systems may, therefore, be able to integrate models from recent dynamic work [13, 14], physical underlying constraints [46, 23], and make use of multi-modal data [109] to solve these challenging tasks. More broadly, the model remains tenable as a core backbone for a 4D world model; not only to reconstruct a single moment but to continuously stream, predict, and alter the 3D scenes—bridging dense image understanding and robot autonomy.

## 7.2 Robotics, Autonomous Navigation, and Physical Simulation

The integration of 3D Gaussian Splatting (3DGS) into robotics and physical simulation represents a paradigm shift from passive scene reconstruction toward actionable, closed-loop perception-action systems. While the previous subsection established 3DGS as a powerful tool for dense visual perception in SLAM, this subsection examines how the explicit, point-based structure of Gaussian primitives—which enables direct manipulation and analysis—positions them as an ideal substrate for physical simulation and robotic interaction. Unlike implicit neural radiance fields that require expensive volumetric sampling for each query, the explicit formulation of Gaussian kernels allows for seamless integration with physics engines, turning the representation from a purely appearance-based geometry into a unified framework for both rendering and predictive dynamics. This expansion from visual perception to physical AI naturally complements the trajectory of Gaussian-based SLAM established earlier, while providing the structured, manipulable scene

substrate that the semantic and language grounding discussed in the following subsection will build upon to deliver interactive and intelligent 3D systems.

The integration of physics into Gaussian representations has been explored through a unified framework where 3D Gaussian kernels are equipped with physically meaningful kinematic deformation and mechanical stress attributes, evolving in accordance with continuum mechanics principles. Notably, in [33], the Material Point Method is seamlessly integrated with the Gaussian representation such that the same primitives serve as the discrete representation for both physical simulation and visual rendering. This exemplifies the principle of *what you see is what you simulate*, negating the need for triangle meshing, marching cubes, or geometric embedding, and enabling high-quality synthesis of novel motions for a variety of materials including elastic entities, metals, non-Newtonian fluids, and granular materials. This tight coupling between rendering and simulation is also reflected in the work on Gaussian Splashing [23], which integrates position-based dynamics with 3D Gaussian kernels to synthesize dynamic fluids. By enhancing the Gaussian kernels with surface normals to refine particle-based simulations, this approach eliminates spiky artifacts and enables correct physical cues for fluid behavior from novel views. Obtaining realistic physical descriptions of elastic objects with Spring-Mass 3D Gaussians [110] similarly integrates a 3D Spring-Mass model into the Gaussian kernel, allowing for the reconstruction, from multiple-view videos, of the visual appearance, shape, and physical dynamics of elastic objects, offering accurate prediction of future deformation states based on different initial configurations.

Beyond simulation, the explicit structure of Gaussian Splatting is a supportive substrate for robotics perception. A novel dual Gaussian-Particle representation was developed to model the physical world in a real-time correctable manner, combining the geometric support of particles with visual rendering of attached 3D Gaussians [25]. This world model allows predictive physics simulation while simultaneously performing online corrections from image observations. By comparing predicted and observed renders, the system computes corrective visual forces that guide particle positions to adhere to physical constraints for continuous real-time synchronization with the environment while remaining consistent with the underlying physics. Operating at 30Hz using only three cameras, this system demonstrates the unique capability of 3DGS to model static, photorealistic scenes while also serving as a policy backbone for embodied agents interacting with environments. The potential of Gaussian-based world models for large-scale autonomous systems is supported by work on very large scenes, which presents a hierarchy of 3D Gaussians that allows real-time rendering while preserving visual quality, with capture trajectories of up to several kilometers [30]. This viewpoint makes 3DGS an attractive component in a real-time rendering stack that requires access to compact and faithful representations.

When used for a higher-level objective, such as robotic manipulation, the explicit nature of the Gaussians allows one to directly segment points and visual semantics relevant to grasping, path-planning, and interaction primitives. Recent work extends 3DGS with a compact Identity Encoding, allowing the Gaussian primitives to be partitioned, tracked, and edited according to arbitrary levels of detail [11]. This enables flexible object removal, inpainting, colorization, and re-composition via user-defined 2D masks, establishing an axiomatic link among geometric structure, visual appearance, and editable semantics—an architecture that is particularly advantageous for fine-grained scene manipulation and object-level reasoning. Moreover, dynamic reconstruction frameworks such as MoSca [111] couple a motion scaffold with an underlying Gaussian scene representation, providing robust correspondences between video evidence and the motion scaffold over time, which are essential for accurate robotic perception in human-centric unconstrained environments. Collectively, these works show that 3DGS emerges as a key enabler for the next generation of 3D world simulators — the basis for generative formats of physical AI, where the ability to virtually interact, physically simulate, and semantically reason over a scene is as important as the photometric fidelity. While the static scene assumption persists in several SLAM systems described previously, the integration of physics and dynamic modeling presented here

overcomes these constraints, unlocking the path forward to the seamless injection of semantic features and linguistic reasoning into dynamic and interactive Gaussian world models that the following section will describe. Rather than remaining an appearance-centric geometry representation, Gaussian Splatting is thus expanding into the broader discipline of 4D world modeling, where continuous simulation and real-time correctability create the backbone for both robot autonomy and human-centric interactive applications.

### 7.3 Language-Embedded and Semantic Scene Understanding

The integration of language and semantic understanding into 3D Gaussian Splatting (3DGS) represents a paradigm shift from purely photometric scene reconstruction toward holistic, queryable 3D representations. Unlike implicit neural radiance fields, whose continuous nature complicates feature attachment, the explicit, discrete structure of Gaussian primitives offers a natural substrate for embedding dense, multi-modal semantic features. This subsection critically examines the methodologies for distilling vision-language features into 3DGS, exploring the mechanisms of feature embedding, representation strategies for scalability, and the synergistic integration of large language models (LLMs) for interactive reasoning. The overarching goal is to move beyond mere object identification towards a comprehensive understanding of *what* objects are and *what they are*, enabling open-vocabulary querying, fine-grained segmentation, and grounded reasoning within the reconstructed scene.

The dominant paradigm for language grounding is semantic feature distillation, which transfers high-dimensional features from pre-trained vision-language models (VLMs) into the 3D Gaussian field. The foundational challenge here is reconciling the spatial and channel inconsistencies between high-resolution RGB images and the lower-resolution, often spatially coarse, feature maps from models like CLIP or SAM. Techniques such as Feature 3DGS [103] have introduced architectural and training changes to efficiently distill features from state-of-the-art 2D foundation models directly into the per-Gaussian attributes. This enables a suite of downstream tasks, including novel view semantic segmentation, language-guided editing, and "Segment Anything" through explicit prompting. The choice of VLM significantly influences the semantic quality: CLIP-derived features offer excellent textual alignment suitable for open-vocabulary classification, while SAM-derived features provide superior instance-level boundaries for precise segmentation. This representation, however, is not without trade-offs. High-dimensional feature vectors drastically increase the memory footprint per primitive, and naïve concatenation can lead to color-feature interference. Subsequent works have sought to mitigate this via compact, learned feature decoders that map lower-dimensional representations to the target space, balancing fidelity against memory consumption.

Beyond distillation, a significant thrust is transitioning from scene-specific models to generalizable, open-vocabulary semantic fields. Rather than overfitting to a single scene, these methods aim to create models that can understand semantics from any new 3DGS representation at training or inference time. This direction tackles the challenge of semantic taxonomy, departing from the fixed category labels of closed-set benchmarks. For instance, comprehensive benchmarks are emerging to evaluate and train models that not only classify objects but also predict fine-grained linguistic attributes—affordances, materials, and function [26, 29]. This highlights a movement toward comprehensive semantic understanding, where the 3D representation is becoming a robust substrate for grounding abstract knowledge in a spatial context.

More recent research has been breaking down the boundaries between the explicit Gaussian primitives, LLMs, and robotic or interactive systems to achieve targeted scene understanding. By integrating a 3D Gaussian field with a Large Language Model, we can deploy contextual-aware canonical phrases to realize zero-shot segmentation. A key innovation is the introduction of *referent tokens*, which are learnable query vectors that bridge the 3DGS representation, the LLM, and lower-level vision models. This setup improves the robust grounding of objects from ambiguous natural language. This level of integration allows the system to understand the

spatial reasoning of 3D scenes. This extends to the domain of interactive robot perception, where the "semantically anchored" reasoning leads to more complex reasoning and text-grounded manipulation capabilities.

Ultimately, semantic embedding represents a foundational step toward creating a unified, interactive framework. The choice of the Gaussian kernel model for this underlying semantic scene is not simply a base but a structured representation that may interact with neural networks [60]. By distilling language features and connecting them to LLMs, those techniques enable editing and parameterization of the environment. This creates a 3D scene that can be queried, segmented, verbalized, and even transformed using high-level linguistic descriptions, which overcomes the bottleneck of static photometry, and steps into the era of machine comprehension of three-dimensional environments.

## 7.4 VR/AR, Interactive Editing, and Real-Time Visualization

The convergence of 3D Gaussian Splatting (3DGS) with immersive technologies marks a paradigm shift in how virtual and augmented reality (VR/AR) environments are captured, transmitted, and interacted with. Building directly upon the semantic embedding and language-grounded querying capabilities introduced earlier, the explicit, point-based nature of the representation provides a unique bridge between photorealistic fidelity and the strict computational, memory, and latency budgets inherent to head-mounted displays (HMDs) and edge devices—transforming the now-queryable 3D scene into a manipulable and interactive medium. In conventional VR/AR pipelines, the creation of interactive scenes is bifurcated: one path prioritizes high-fidelity offline rendering at the cost of interactivity, while the other favors lightweight, but visually simplistic, polygon-based models. 3DGS directly addresses this dichotomy, extending the semantic queryability from the prior section into a dual role: it serves simultaneously as a high-fidelity radiance field, a manipulable point cloud, and a rasterization-ready primitive set, enabling streaming detail and in-place, language-guided editing without compromising frame rate—functional demands that arise naturally from the semantic understanding achieved through feature distillation and LLM grounding established above.

A primary challenge for content streaming arises from transmission bandwidth, particularly when moving from the single-scene, language-anchored representations discussed previously to distributed and remote rendering systems. A single, overfitted scene can contain millions of Gaussian primitives, representing hundreds of megabytes of data, rendering it impractical for standard wireless transmission to the resource-constrained HMDs. Contemporary methodologies tackle this through simplified, device-adaptive paradigms that maintain the semantic cohesiveness introduced earlier. Approaches such as Mini-Splatting [19] reorganize the spatial distribution of Gaussians via blur splitting and depth reinitialization to represent scenes with a constrained primitive count. This is complemented by level-of-detail (LoD) structures that allow the culling of far-field primitives while preserving the dense feature fields required for open-vocabulary understanding on-edge. Crucially, these compression strategies must operate under a rate-distortion trade-off specific to remote rendering scenarios, where the goal is not merely to reduce file size but to reduce the *per-frame* bitrate delivered to the GPU. Compression via spectral pruning and neural compensation [73] demonstrates that a lightweight MLP network head can mix a primal, selected set of splatted Gaussian features to compensate for quality losses, creating a linked representation where the "latent" state of the field—the true mapping between primitives and the MLP—can be compressed for transmission, ensuring that high-level semantic grounding is preserved without exhausting bandwidth budgets.

Beyond static fidelity, this tight coupling of the representation to interactive physics and direct manipulation resolves historical challenges of editing digital content. Where NeRF-based offline rendering requires significant compute per update, 3DGS facilitates direct parameter editing of geometric primitives—a capability enhanced by the semantic anchoring from the previous section, which provides practical affordances for real-time aesthetic changes to object

translation, removal, and transfer [112], as well as feature-level manipulation to achieve novel effects [103]. The explicit query-ability established via the language-aware features extends seamlessly into immersive interactions. By distilling semantic features from 2D foundation models (e.g., CLIP, SAM) into these volumes, semantic fields are established, enabling direct manipulation of objects sampled by a single point or bounding box [103]. This capability integrates smoothly with diffusion models, where natural language instructions can be used to iteratively edit the visual appearance of the whole scene, transporting the high-level semantic control from the prior section into a visual and immersive arena, and ushering in a new era of casual user interaction with the 3D world, bypassing specialized graphics tools for semantic and text-guided pipelines [67, 100].

This realism and robust interactivity gain further traction when considering the distantly handled case of low-quality, insufficiently textured inputs. Realistic and robust interaction within the reconstruction of broader, unbounded scenes depends on precise geometry. On real-time mobile devices, the lack of accurate geometry initialization severely degrades optimization. This is addressed by integrating Structure-from-Motion (SfM) priors within Gaussian-based SLAM frameworks, where the robust sparse initialization from SfM is coupled with the inherent strength of the primitives for local reconstruction—preserving the fidelity required for tracking and interaction. Furthermore, introducing optical flow awareness into the splatting representation—as done by [113]—improves geometry for complex or sparse motion cases, which is crucial for the interplay between real-time tracking and low-memory consumption on edge devices. Despite these advances, the bottleneck of splatting-based immersive rendering remains asynchronous I/O and texture utilization, where mobile devices experience compression and latency in the suboptimal graphics pipeline of HMDs. The explicit representation’s strength in ease of manipulation while facilitating interactive language-guided and physics-driven editing as shown above—exists within a context that necessitates new forms of constrained transmission and sparse parameter transport to maintain a minimum latency loop that satisfies the real-time demands of VR/AR and the computational suite of autonomous systems discussed in the following subsection.

The path forward lies in the synthesis of physical priors with rate-aware primitives, fostering a natural transition from the semantic interactivity of this section to the highly specialized, scaled applications of the next. We anticipate a trend where head-mounted 3DGS will be coupled with dynamic scene decomposition as perfected in the following section’s autonomous driving and event-driven and neuromorphic rendering, shifting focus from "universal load" to "sparse or local load" editing. The promise of 3DGS in this domain is double-sided: the point-based representation becomes a bridge, being at once the medium for interactive physics simulations and the carrier of global illumination. Interactive networks in future architectures must be trained to compress the 3D hierarchy, with direct access from interacting modules for 3D dynamic editing, rather than solely accessing the viewing parameters, offering a competitive new cornerstone for communication and interaction with the scene format. This sets the stage for the re-evaluation of the core assumptions of static geometry and optical fidelity, as fundamentally distinct, high-stakes, transformative and demanding applications (urban modeling, surgical robotics, and tomographic vision) push the framework to incorporate robust 4D dynamics and physical plausibility, as the following subsection dissects in depth.

## 7.5 Specialised Domains and Scientific Imaging (Autonomous Driving, Medical, and Tomography)

The deployment of 3D Gaussian Splatting (3DGS) in specialised, high-impact domains necessitates a fundamental re-evaluation of its core assumptions—namely, static geometry, optically transparent media, and a well-conditioned camera model. While the base representation excels in object-centric and bounded scenes, autonomous driving, surgical robotics, and tomographic imaging present unique constraints ranging from extreme scale and dynamic object decomposition to the physics of non-visible radiation and highly deformable tissues. This subsection explores how the



core 3DGS framework is being systematically adapted to satisfy the accuracy, robustness, and real-time requirements of these demanding application areas.

A primary challenge is modeling unbounded, dynamic urban environments from a monocular rig. An early representation, Street Gaussians [114], addressed this by decomposing the scene into a set of point clouds with semantic logits and 3D Gaussians, each associated with either a foreground vehicle or the static background. This explicit decomposition allows for the learning of optimizable tracked poses and dynamic appearance models for each object, enabling photorealistic novel view synthesis and scene editing operations with high quality. This segmentation is fundamental for creating realistic and interactive simulations. Building upon this, HUGS [115] formulates a joint optimization problem for geometry, appearance, semantics, and motion over a combination of static and dynamic Gaussians, producing a holistic urban representation that facilitates downstream tasks like semantic segmentation and object tracking. The efficacy of these models, however, is often limited by their reliance on accurate tracked object poses, motivating continued research into more robust, self-supervised 4D decompositions. These advances are the pivotal step for autonomous driving, enabling not only photorealistic sensor simulation but also a deeper semantic and geometric understanding of the surrounding world for navigation and planning.

In the medical sphere, the limitations of structure-from-motion in texture-poor environments dictate bespoke solutions. Given that endoscopic scenes in minimally invasive surgery are highly deformable, have significant specularities, and lack reliable SfM initialization, methods are explicitly designed to circumvent these failures. LGS [116] provides a representation designed to meet the real-time and compute constraints of surgical environments. By integrating deformation-aware pruning and feature field condensation, it minimizes redundant Gaussian quantities and attributes, thereby enabling a reduction in memory consumption while maintaining high visual fidelity for deformable tissue reconstruction. Further, in the limited frame-rate computation and reliance on monocular captures from a single endoscope, the ability of splatting-based methods to render only relevant components without sacrificing speed is critical. These methods serve as a powerful testament to the adaptability of the representation when the prior constraints of static cameras and sufficient texture are removed, signifying a promising avenue for guidance systems and surgical training in robotic surgery.

Beyond optical sensing, a more radical shift entails adapting the Gaussian kernel to the physics of invisible radiation, underlining its versatility beyond conventional imaging. The complexities inherent in medical image data (such as background noise and distortions) can be addressed by this representation. For domains like precise and inverse rendering, the challenge lies in direct reconstruction of volumetric data and, crucially, the ability to efficiently understand the underlying data generation process. Since the problem consists of recovering 3D volumes from 2D projections and performing reconstruction in complex, high-motion scenarios, the conventional assumptions fall short. Instead, these methods use a bespoke approach (e.g., Gaussian splatting with a unique formulation) that cannot be collapsed into shallow multi-layer perceptrons and optimizations. They require the regrouping of a set of Gaussian kernels and the deformable-body structure of the scene to get a faithful solution that can be rendered in real-time. The generative nature of this representation allows for rendering to run at interactive rates, but the principal factors are the design and implementation of the underlying differentiable rasterization pipeline and the capacity to be tuned for different physics.

Across these disparate applications, a critical synthesis emerges: the Gaussian primitives at the core of 3DGS are not simply a rendering tool but a versatile, differentiable data structure. Their explicit nature enables the 3D reconstructions and semantic compositions, providing an efficient alternative to NeRF-based representations. Looking forward, we expect a convergence of these methodologies. For example, techniques from medical domain—such as handling unreliable poses and deformable objects—are directly transferable to autonomous driving in complex, dynamic traffic situations, while physics-integrated, efficiency-first Gaussian representation will

see a more pronounced focus on geometric fidelity and topological consistency. This frontier is not just about extending an algorithm to new tasks but solidifying the shift of 3DGS from a tool for image synthesis to a core representational foundation for 3D scene understanding and dynamic modelling across scientific and engineering practice.

## 8 Conclusion and Future Research Directions

This survey has charted the rapid evolution of 3D Gaussian Splatting (3DGS) from a novel differentiable rendering algorithm [7] to a foundational technology driving innovation across computer graphics, vision, and robotics. As we conclude, we synthesize the technological progress that has positioned 3DGS as a cornerstone of modern scene representation while identifying the complex challenges and promising opportunities that lie ahead. The journey from point-based rendering [5, 28, 6] to sparse voxel grids [1, 3] and coordinate-based networks [2] has cumulated in a representation that offers an unprecedented combination of photorealistic rendering speed, editability, and adaptability. The methodological advances in adaptive density control [41, 19], regularization [45, 46, 60, 46], and compression [27, 66, 72] have collectively expanded the capacity and robustness of the base framework, enabling applications as diverse as surface mesh extraction [117, 37, 69] and high-frequency appearance modeling [45, 65]. This synthesis reveals a field in a state of healthy, rapid maturation, yet several deep-seated questions remain.

A critical disconnect persists between the empirical efficacy of 3DGS and a mature theoretical understanding of its behavior. While the representational capacity of millions of anisotropic Gaussians is intuitively clear [7], a formal framework for analyzing its expressiveness, generalization bounds, and spectral properties remains absent. This lack of theory impedes principled, rather than heuristic, solutions to open problems. For example, the canonical issues of early over-reconstruction, gradient collisions [41], and the suboptimal spatial distribution of Gaussians [19] have all been addressed via empirical, albeit effective, heuristics. Similarly, SfM initialization [45]—a brittle dependency—and the manifold difficulties in handling sparse or degraded inputs [45, 65] highlight the need for a more principled approach to optimization and regularization. As the representation scales from single objects to unbounded, city-scale scenes [9, 114], the coherence of these point clouds will become an even more pressing issue. We posit that future research must move beyond empirical refinement and begin to develop a statistical understanding of the representation to guide densification, constraint, and regularization algorithms in a more systematic manner.

A set of significant challenge is the robustness of 3DGS in the face of unconstrained, real-world conditions. While the base model excels with well-conditioned, dense-capture inputs, its performance degrades when the input is sparse [45], blurry [65], or contains transient occluders [93]. The success of prior-based methods, particularly those integrating monocular depth and normal priors [46], and generative models for inpainting [99], demonstrates a clear direction: integrating large-scale training data with the optimization process. Future frameworks should seamlessly fuse data-driven priors with the physics-based, per-instance optimization of Gaussian splats to enforce global semantic and photometric consistency. The coupled problem of geometry and appearance optimization is particularly acute; the “Murphys’ law” of inverse specific rendering—where the optimizer might explain specular reflections as a textured surface—remains a fundamental challenge for 3D structure from images. Achieving robust and physically faithful decomposition of geometry, material, and lighting [45, 46, 118] is not just a performance issue; it is a prerequisite for the practical deployment of 3DGS in demanding fields such as facial digital twins, embedded digital twins, and the digital reconstruction of virtual environments.

The performance bottleneck has gradually shifted from rendering speed to the front-end of storage and training cost. While rendering has reached a speed of 700+ FPS for static scenes, the storage and memory footprint of millions of Gaussians remains grossly disproportionate. Compression methods [66, 27, 119] provide efficient codecs, but they often trade off quality

or require an expensive offline optimization. A unifying perspective is to think of 3DGS as a data type; future work should aim to compress a full further, perhaps by sharing instance-level patterns across scenes or by learning a lower-dimensional latent space. For dynamic scenes, the explosion of data is even more significant. While deformation-based methods [46] provide elegant solutions for continuous motion, they still struggle with sudden, narratively- and structural-changing movements and long sequences. Compact and lossless representations, perhaps through a hierarchical structure for both static and dynamic, will be crucial for practical deployment on resource-constrained edge devices.

Our survey also reveals a glaring standardization gap in the evaluation of performance benchmarks. The community’s over-reliance on synthetic datasets like the ones from previous work [1, 3] and a handful of object-centric captures like the [1] does not capture the robust fields "in-the-wild" nature. This heterogeneity is often selectively reported, leading to a fragile underst of the field. We advocate for a comprehensive evaluation on a split of scenes, separating "geometry-based" and "appearance-based" tasks, and further include a 'cross-dataset' split to truly assess generalization. These benchmarks should consider and incorporate uncertainty, such that a model that is confident in a prediction is prized, rather than a model that is certain. Unfair and naive training-vs-quality trade-offs, where one method is trained on a curated high-res dataset and the other on a video of first-lifetime, are a challenge that must be avoided.

The future of 3DGS lies in bridging its explicit geometry with the semantic and physical world. The field is trending toward a multi-layered world where geometry, semantics [11] and physical law coexist in a holistic representation. We envision a framework where 3DGS does not exist as a unique representation but as a core "rendering-and-interaction" layer that can be trained with differentiable physics, with language for grounding, and with actions for intelligent, real-time system. The next step will be toward 3D foundations—where Gaussians are not just a goal in themselves but a means to an end, deployed across modalities to analyze, simulate, and predict physical world dynamics. The integration of physically embodied agents [46] and human-centered avatars [24] within the Gaussian framework represents a frontier for this representation, enabling a new generation of interactive 4D world models. More than just a rendering engine, 3DGS is becoming a de facto computational unit for real-time and interactive world from multi-modal inputs. As the field advances and it makes a profound impact on computer vision, it will also bring along with it questions about efficiency, robustness, and governance, and it is our hope that this survey has a graph to the way.

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